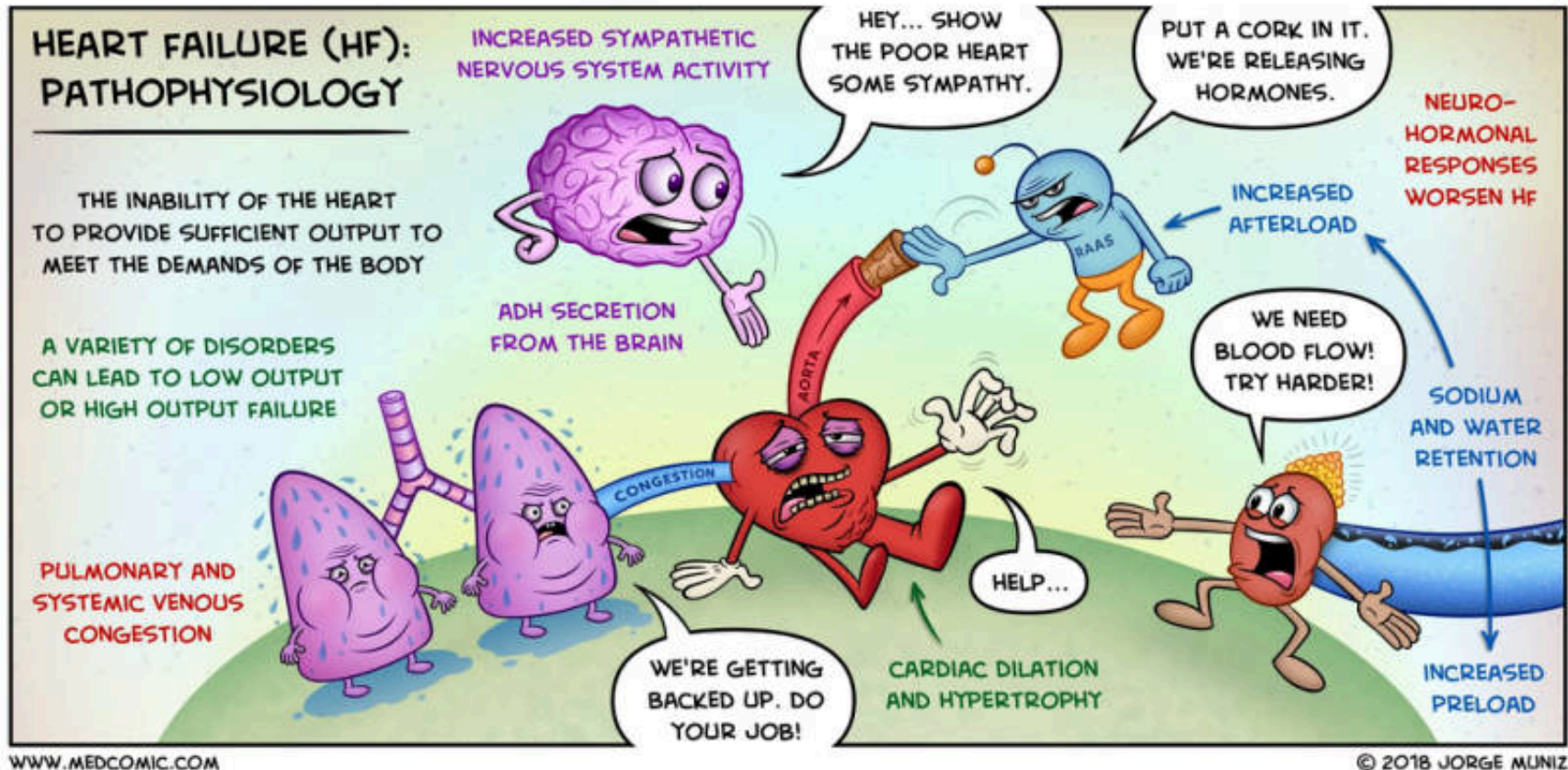


Pharmacology of Heart Failure



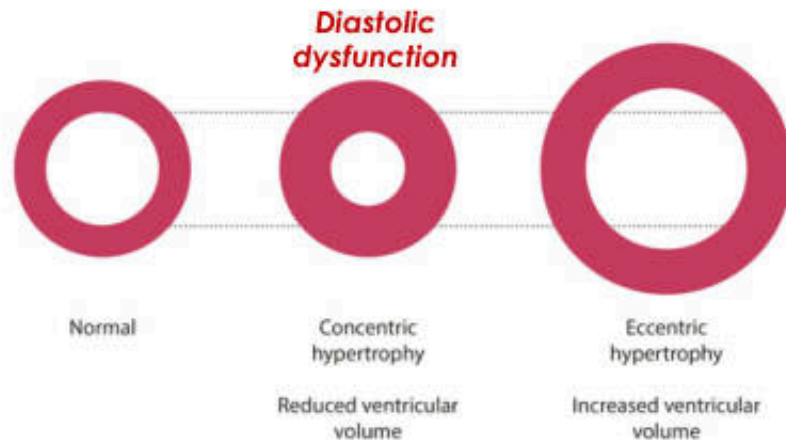
Anunchai Assawamakin, B.Sc.(Pharm), Ph.D.

Department of Pharmacology, Faculty of Pharmacy, Mahidol University

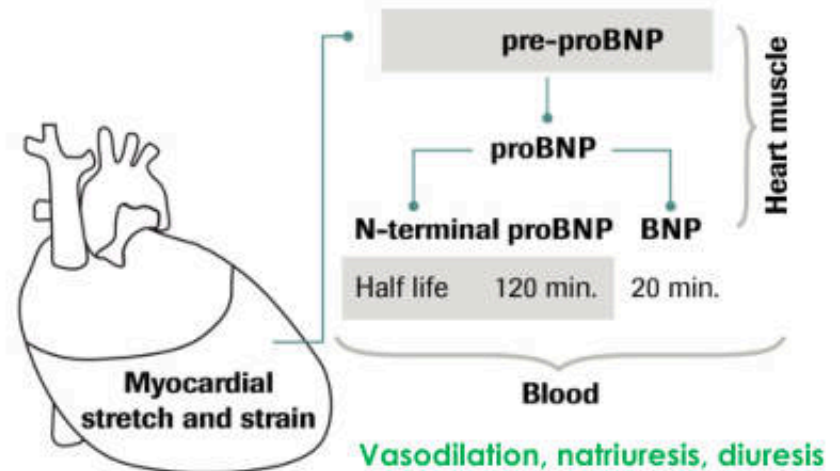
Heart Failure (HF)

- HF is a **clinical syndrome** with **current** or **prior**
 - Symptoms and signs caused by **structural/functional cardiac change**
- And collaborated by at least one of the following:
 - Elevated **natriuretic peptide** level
 - Objective evidence of **cardiogenic pulmonary** or **systemic congestion**

Types of hypertrophy

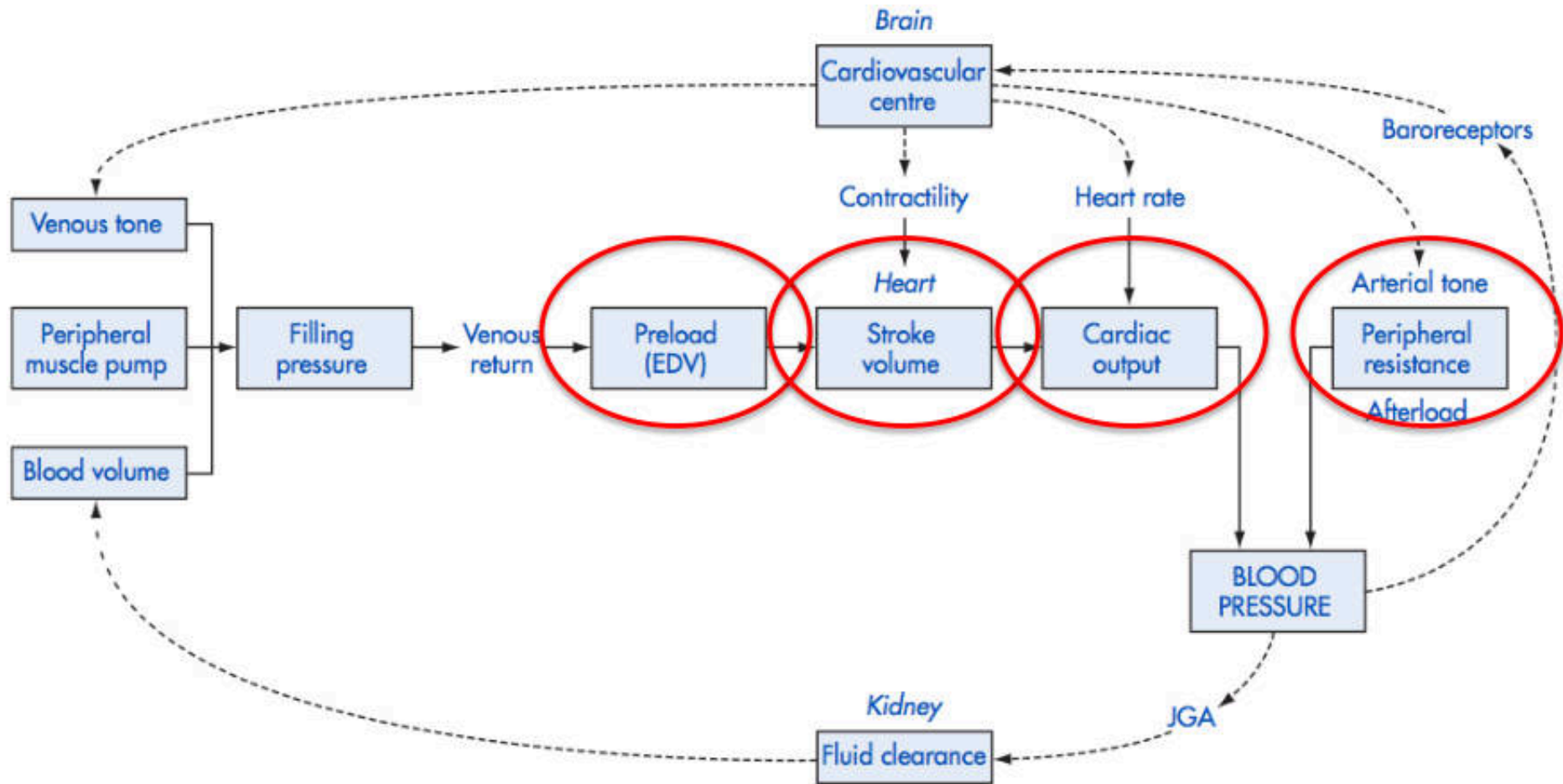


Biomarker for HF: BNP & NT-Pro BNP



Heart Failure

- Heart failure = **inadequate blood supply** to the body which manifests in various **symptoms from tissue hypoxia**
 - Fatigue, weakness
 - Dyspnea on exertion, orthopnea
 - Fluid retention; peripheral edema, pitting edema
- Goal of Treatment
 - Improve patient symptoms
 - Slow/reverse deterioration of myocardial function
 - **Reduce mortality**



Physiology

- Cardiac output (CO) = Stroke volume (SV) x Heart rate (HR)
- Pressure = Flow x Resistance
 - Mean Arterial Pressure (MAP) = CO x Systemic vascular resistance
- Factors affecting cardiac output:
 - Chronotropy (Heart rate)
 - Inotropy (Contractility)
 - Preload (Intravascular volume)
 - Afterload (Systemic vascular resistance)

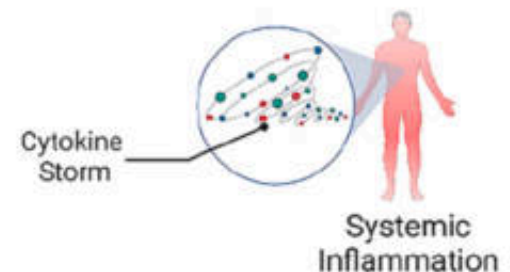
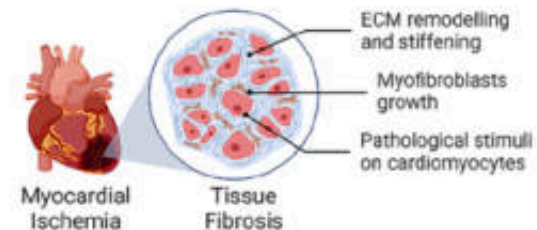
Pathophysiology of HFrEF (Heart Failure with reduced Ejection Fraction)

- HFrEF is predominantly a **systolic dysfunction** disorder, where the heart's ability to contract and pump blood effectively is compromised.
- The ejection fraction is reduced, typically below 40%, due to intrinsic damage to cardiomyocytes and structural remodeling of the heart.
- **Mechanisms in HFrEF**
 - **Myocyte Loss and Remodeling**: Loss of heart muscle cells leads to an enlarged, weakened heart.
 - **Neurohormonal Activation**: Overactive nervous and hormonal systems place more stress on the heart.
 - **Calcium Dysregulation**: Abnormal calcium handling reduces the strength of heart contractions.

Summary: HFrEF focuses on problems with the heart's ability to pump, lowering blood ejection.

Pathophysiology of HFpEF (Heart Failure with preserved Ejection Fraction)

- HFpEF is primarily characterized by **diastolic dysfunction**, where the left ventricle (LV) has difficulty relaxing and filling, despite preserved contractility (ejection fraction).
- **Mechanisms in HFpEF**
 - **Myocardial Stiffness**: Increased stiffness from collagen and fibrosis accumulation.
 - **Endothelial Dysfunction**: Impaired blood vessel function makes it harder for blood to return to the heart.
 - **Systemic Inflammation**: Chronic inflammation from comorbidities (e.g., diabetes, hypertension) increases myocardial stiffness.



Summary: HFpEF focuses on issues with the heart's relaxation, reducing its ability to fill with blood effectively.

Acute Heart Failure (AHF): Standards of Treatment

Stratification of Pts With AHF

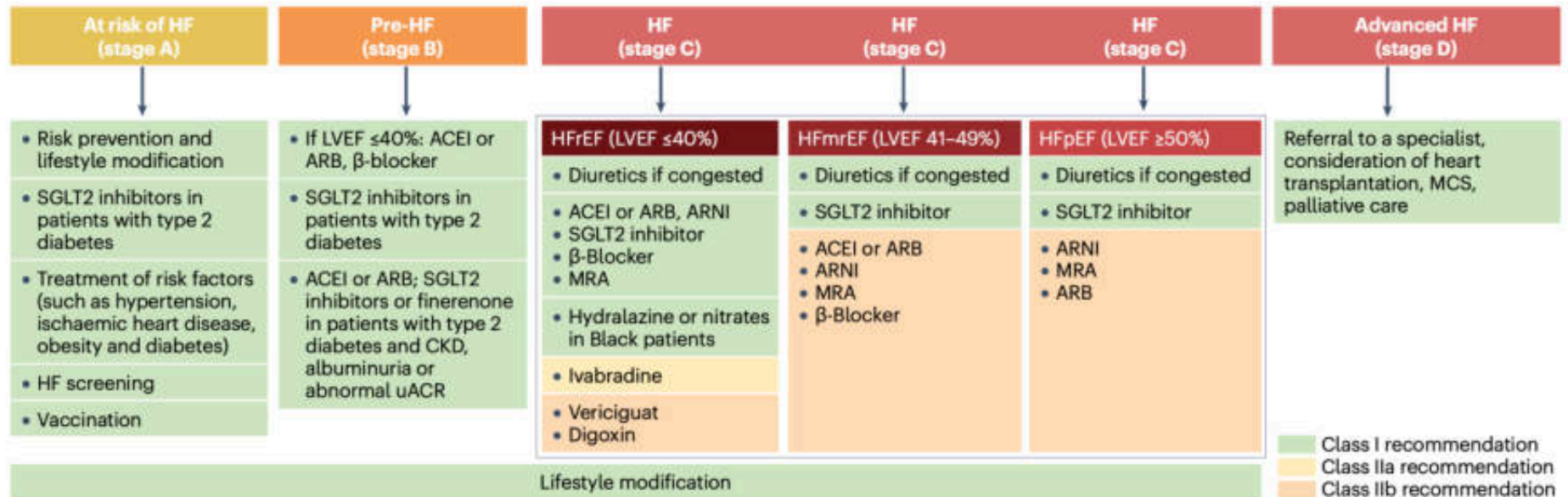
Congestion at Rest?

(eg, orthopnea, elevated jugular venous pressure, pulmonary rales, S3 gallop, edema)

Low Perfusion at Rest?
(eg, narrow pulse pressure, cool extremities, hypotension)

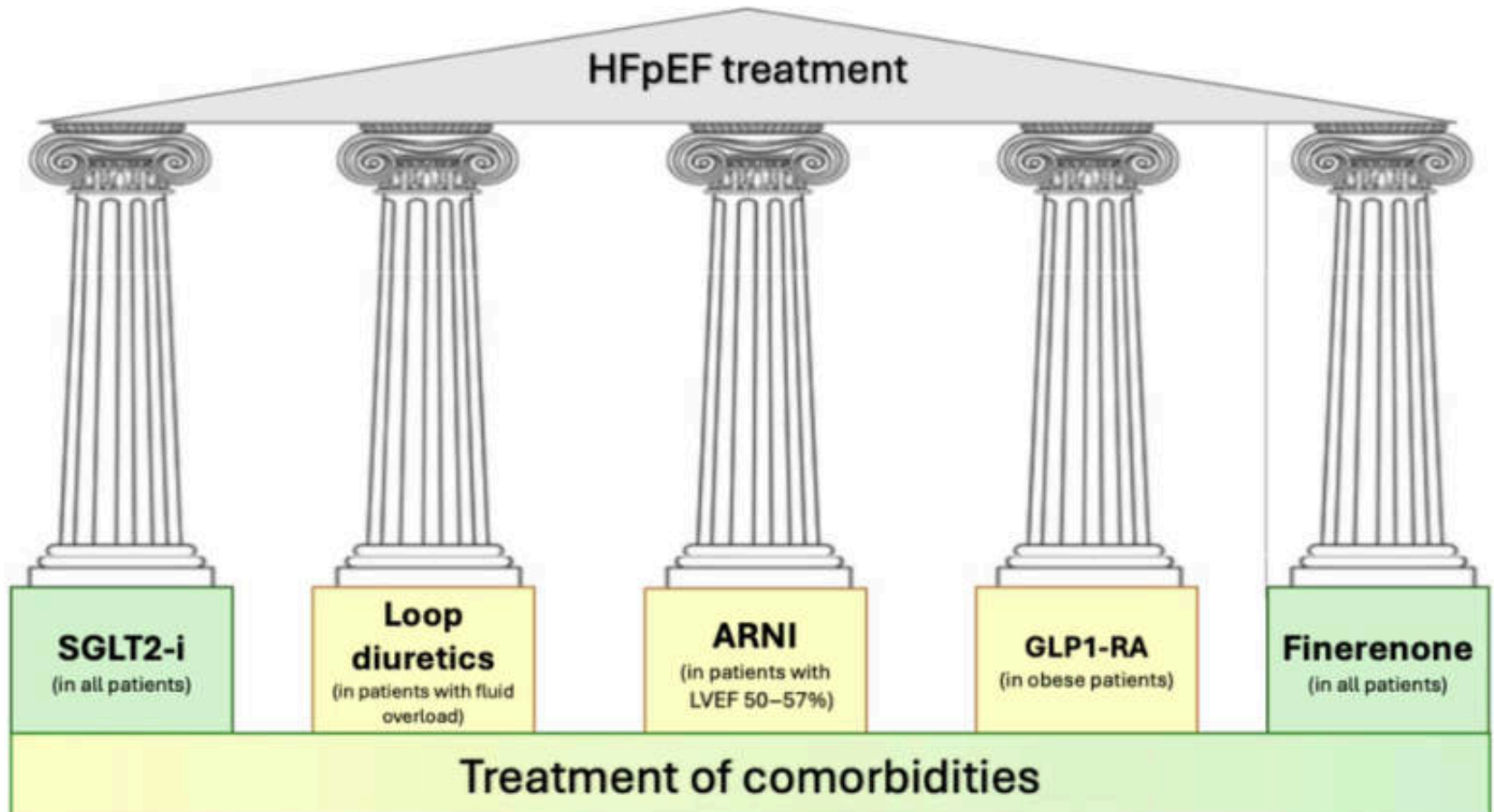
		No	Yes
No	Warm and Dry		Diuretics, vasodilators
	Cold and Dry	Inotropes, vasodilators	Diuretics, inotropes, vasodilators
Yes	Cold and Wet		

Recommendations for pharmacological therapy across all stages of HF

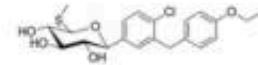
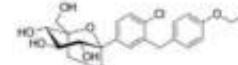
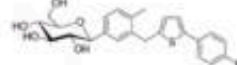
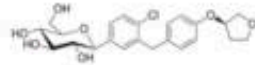
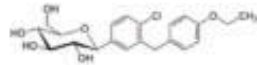


ACEI, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; ARNI, angiotensin receptor–neprilysin inhibitor; CKD, chronic kidney disease; GDMT, guideline-directed medical therapy; HFmrEF, HF with mildly reduced ejection fraction; HFpEF, HF with preserved ejection fraction; HFrEF, HF with reduced ejection fraction; LVEF, left ventricular ejection fraction; MCS, mechanical circulatory support; MRA, mineralocorticoid receptor antagonist; SGLT2, sodium–glucose cotransporter 2; uACR, urine albumin-to-creatinine ratio

HFpEF treatment approach



SGLT2 inhibitors: from glucose-lowering to cardiovascular benefits



GLIFLOZINS: MECHANISTIC INSIGHTS

Pleiotropic effects

Systemic levels

- ↓ Inflammation
- ↓ Oxidative stress
- ↑ Autophagy
- ↑ Mitochondrial function
- ↓ Insulin resistance
- ↓ NHE activity
- ↓ Fibrosis
- ↓ Ferritin
- ↓ Hepcidin

Vasculature

- ↓ Endothelial dysfunction
- ↓ Arterial stiffness
- ↓ Blood pressure
- ↓ Atherogenesis

Liver

- ↑ SIRT1
- ↑ HIF-2α
- ↑ Erythropoietin

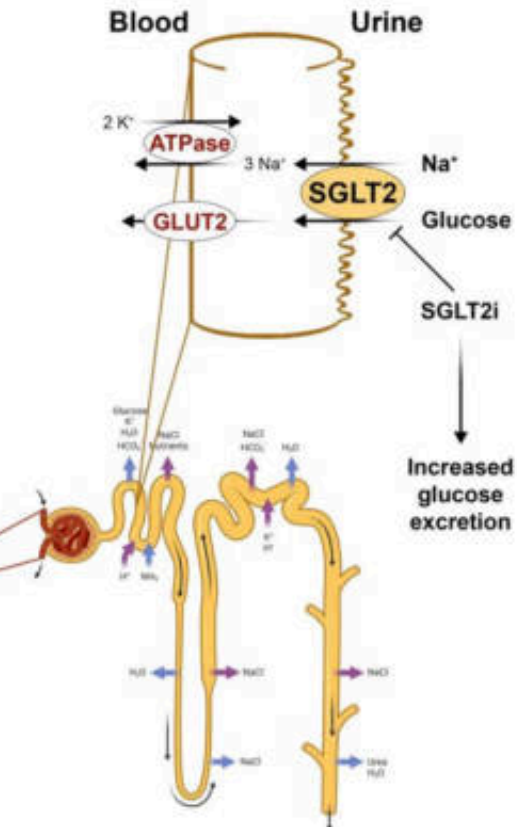
Heart

- ↑ Energetics
- ↓ Epicardial fat
- ↓ Pre- and afterload
- ↑ Ca²⁺ homeostasis
- ↑ Myocardial contractility
- ↑ Ketone body utilization
- ↑ Left-ventricular function

Kidney

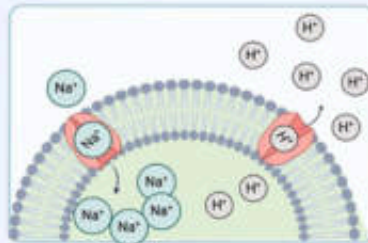
- ↑ Glycosuria
- ↑ Natriuresis
- ↑ Diuresis
- ↓ Albuminuria
- ↓ Glomerular pressure
- ↑ Tubuloglomerular feedback
- ↓ Intrarenal RAAS activation

Glucose-lowering effects



GLIFLOZINS: CARDIAC EFFECTS

Na⁺/H⁺ Exchanger 1



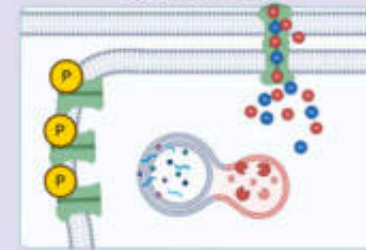
- NHE1 inhibition → cytosolic [Na⁺] ↓; cytosolic [Ca⁺⁺] ↓; coronary vasodilation ↑

Cardiac metabolism



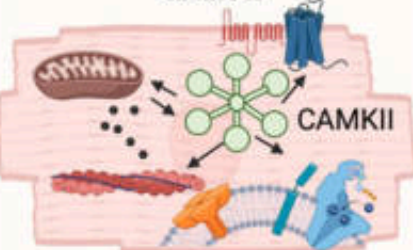
- HF Metabolic shift reversion:
 - glycemia ↓
 - carbohydrate consumption ↓
 - lipid consumption ↑
 - ketone bodies production and consumption ↑
- Myocardial intracellular sodium ↓; phosphocreatine/ATP ratio ↑ → mitochondrial function ↑

Connexins



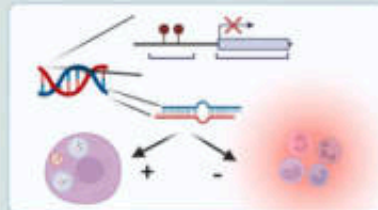
- In diseased cardiomyocytes
 - Cx expression ↑
 - pS368-Cx43 ↑
 - Cx43 lateralization ↓
 - arrhythmic risk ↓
- In healthy cardiomyocytes
 - Cx expression ↓ → Autophagy ↑
 - pS368-Cx43 ↑
 - cardioprotection ↑

Ca²⁺/calmodulin-dependent protein kinase II



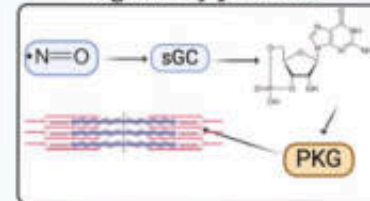
- Mitochondrial function ↑ → ROS ↓ → Camkii ↓ → systolic/ diastolic dysfunction ↓
- TLR4/CaMKII signaling ↓ → left ventricular remodeling, ventricular arrhythmias ↓

Epigenetic modulation



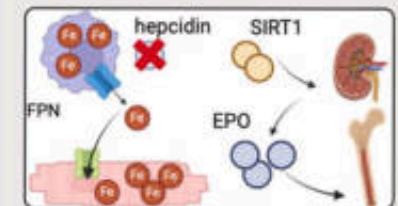
- Sirtuin activation ↑ → HDAC inhibition → autophagy ↑
- beta-hydroxybutyrate ↑ → HDAC inhibition → autophagy ↑
- HDAC2 ↓ → NF-κB and SOD2 ↓

Phosphorylation of myofilament regulatory proteins



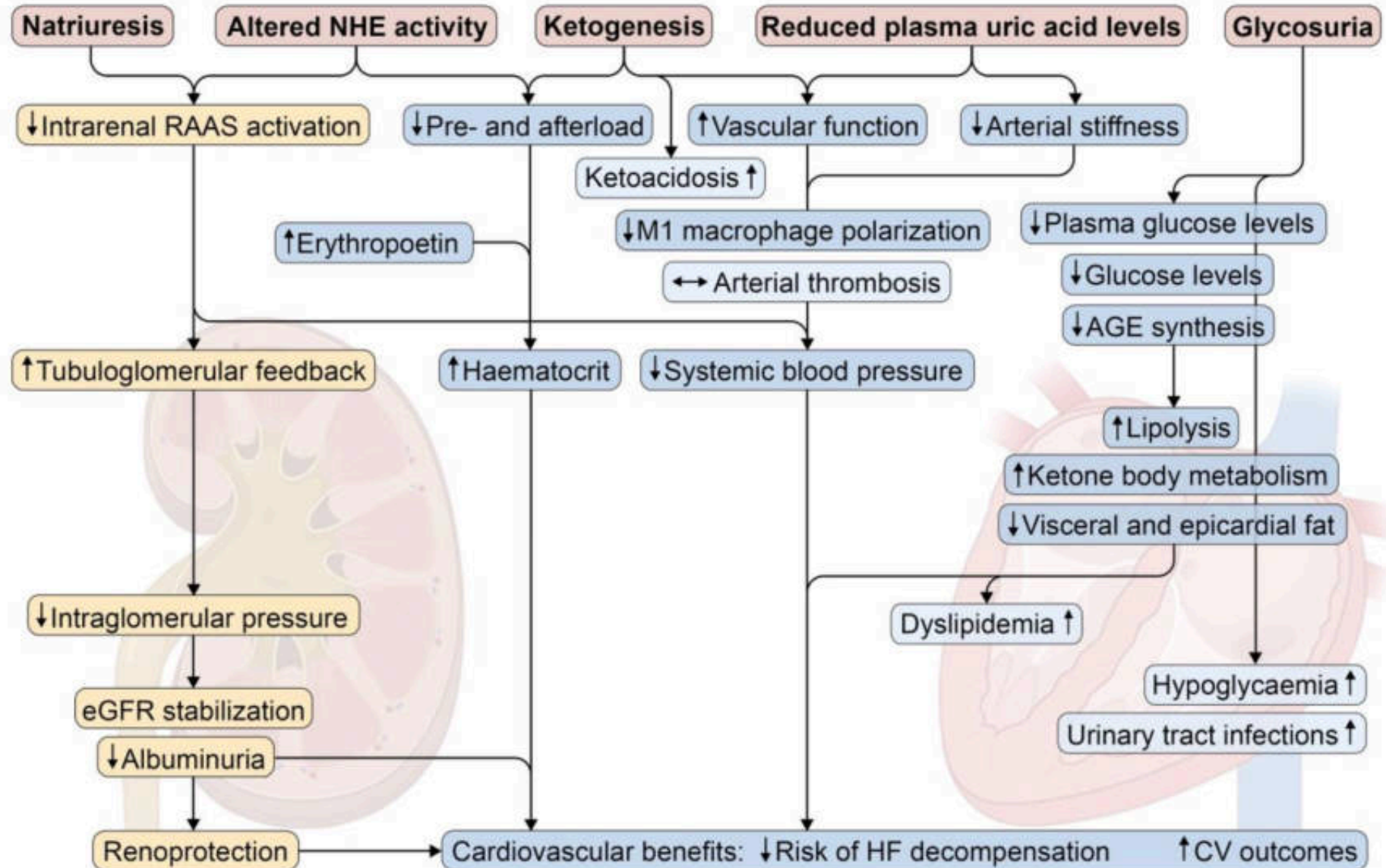
- NO-sGC-PKG pathway ↑ → titin phosphorylation ↑ → passive stiffness ↓

Erythropoiesis and iron metabolism



- Hepcidin ↓ → iron mobilization → iron in cardiomyocytes ↑ → mitochondrial function ↑
- SIRT1 ↑ / peritubular interstitial fibroblasts ↑ → EPO ↑ → hematocrit ↑ → delivery of O₂ ↑

GLIFLOZINS: RENAL AND CARDIOVASCULAR EFFECTS



HFrEF (Heart Failure with Reduced Ejection Fraction)

SGLT2 inhibitors help improve outcomes in HFrEF by:

1. **Reducing Blood Volume and Pressure:** By increasing sodium and water excretion, SGLT2 inhibitors lower blood volume and pressure, reducing heart workload.
2. **Lowering Cardiac Workload:** This helps the weakened heart pump more effectively, easing the burden on the heart muscle.
3. **Decreasing Fibrosis in Heart Tissue:** SGLT2 inhibitors help reduce fibrosis (scar tissue) in heart muscle, improving elasticity and function.
4. **Reducing Inflammation:** The anti-inflammatory effects help prevent further damage to the heart muscle.
5. **Supporting Energy Use:** SGLT2 inhibitors encourage the heart to use ketones, an efficient energy source, enhancing heart function.
6. **Reducing Oxidative Stress:** Lower oxidative stress reduces heart cell damage, improving heart resilience.
7. **Balancing Electrolytes:** By managing sodium and potassium levels, SGLT2 inhibitors reduce the risk of arrhythmias (irregular heartbeats).

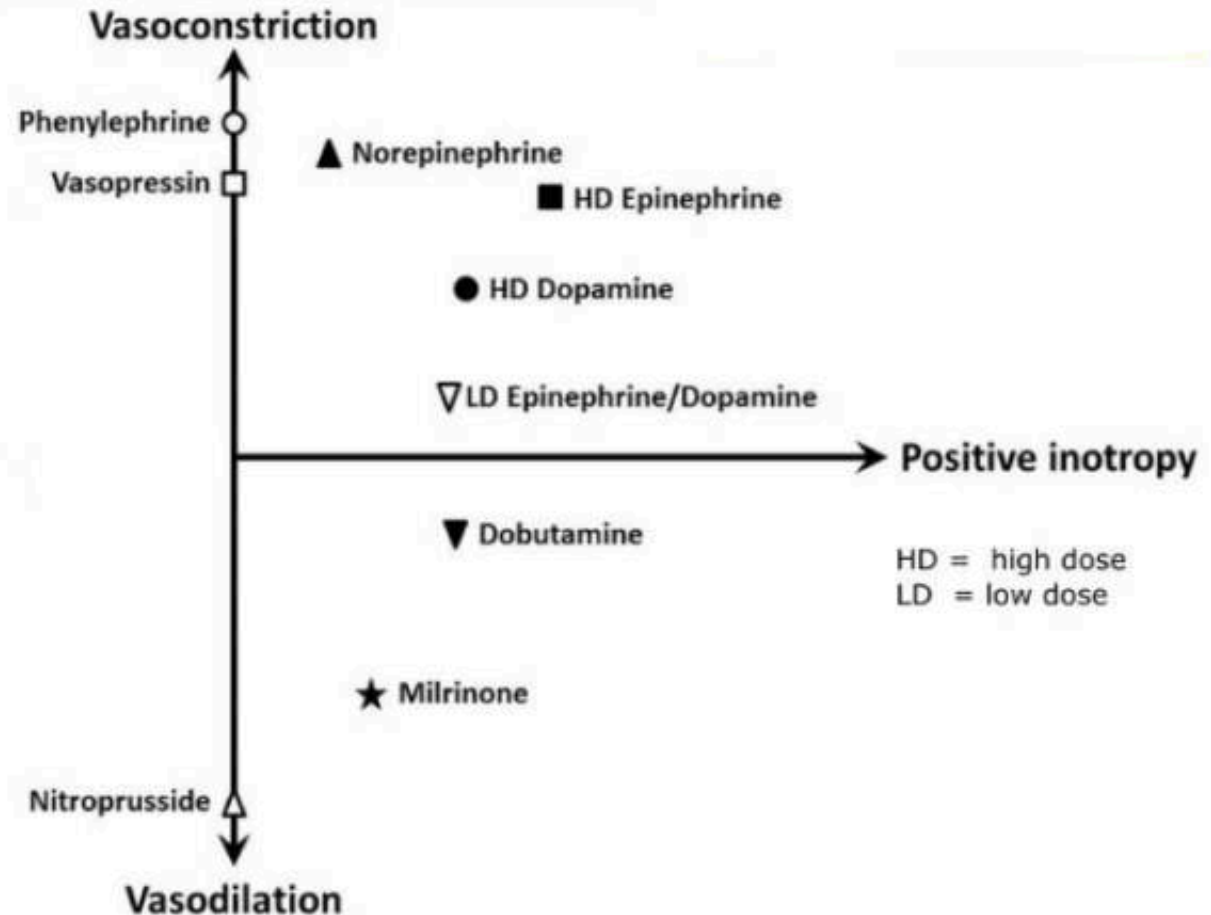
HFpEF (Heart Failure with Preserved Ejection Fraction)

SGLT2 inhibitors help improve outcomes in HFpEF by:

1. **Lowering Blood Volume and Pressure:** This reduces blood pressure and volume load, beneficial for a stiff heart with preserved ejection fraction.
2. **Enhancing Diastolic Function:** Reduced blood volume and pressure help the heart fill more effectively, improving diastolic function.
3. **Reducing Inflammation and Fibrosis:** This reduces stiffness in the heart muscle, supporting its ability to relax and fill with blood.
4. **Decreasing Oxidative Stress:** Lower oxidative stress helps prevent cellular damage, supporting healthier heart muscle.
5. **Promoting Energy Efficiency:** By shifting energy use toward ketones, the heart has a more efficient fuel source, beneficial for a heart struggling to relax.

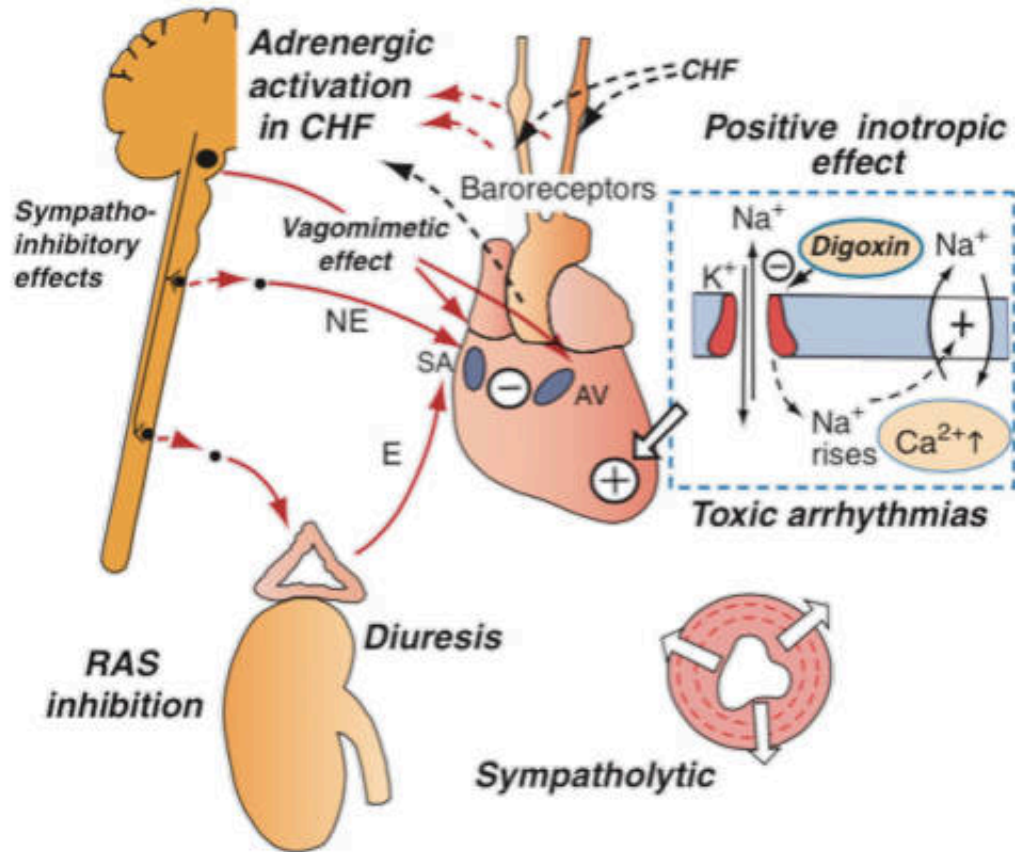
Cardiac Contractility Drugs

- Cardiac glycosides
 - Digitalis/**digoxin**/digitoxin
- β_1 -Adrenergic receptor agonists
 - **Dobutamine**, dopamine, epinephrine, norepinephrine
- Phosphodiesterase inhibitors
 - amrinone, **milrinone**



Digoxin

- Cardiac glycosides inhibit the Na^+/K^+ ATPase pump on cardiac cell membranes



- $\uparrow$ Intracellular $\text{Na}^+ \Rightarrow \downarrow \text{Ca}^{2+}$ exchanged by the $\text{Na}^+/\text{Ca}^{2+}$ exchange pump
- Accumulation of intracellular Ca^{2+}
 - $\uparrow$ Increase cardiac contractility

Digoxin: Clinical Effects

- ↑ Contractility (inotropy)
 - ↑ Stroke volume/Cardiac Output
 - Weak inotrope
- There is mild systemic, coronary arterial, and venous vasoconstriction (especially when given by intravenously)
- There is some diuretic effect as it **inhibits reabsorption of Na^+ at the kidneys.**
- Research shows that use of digoxin has little to no effect on long-term survival.
 - It is not recommended in NYHA Class I/II CHF unless there is atrial fibrillation (AF)

Digoxin

- Indications
 - Congestive Heart Failure (symptomatic HFrEF)
 - Antidysrhythmic agent
 - Used to slow the ventricular rate (< 100 bpm) in atrial flutter/fibrillation
- Availability: IV, oral tabs, soft gel capsules, elixir
- $T_{1/2} = 1.5\text{--}2$ days (easy to build up blood levels)
- Excretion is proportional to glomerular filtration rate (GFR)
- Narrow therapeutic index: need to avoid toxicity

Digoxin: Drug Interactions

▪ **Decrease serum levels**

- Antacids
- Kaopectate
- Bran
- Phenytoin
- Propafenone
- Neomycin
- Sulfamethoxazole
- Phenobarbital

▪ **Increase serum levels**

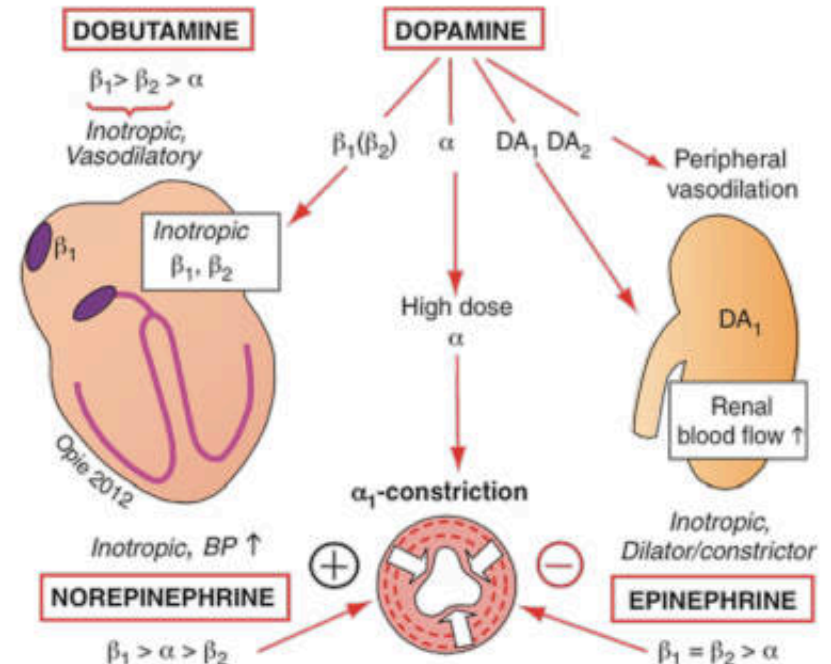
- Quinidine
- Amiodarone
- Flecainide
- Verapamil
- Spironolactone
- Triamterene
- Indomethacin
- Cyclosporine

Digoxin: Adverse Effects

- GI – most common
 - Nausea, vomiting, diarrhea, anorexia
- Cardiac
 - Any abnormal brady- or tachydysrhythmia
 - Exception: atrial fibrillation with rapid ventricular response
- CNS
 - Confusion, visual disturbances (yellow halos around lights)

β -Adrenergic Agonists: Dobutamine

- Selective β_1 agonist
 - $\uparrow$ cAMP $\Rightarrow$ activates protein kinases
 - $\uparrow$ Ca^{++} in cell & sarcoplasmic reticulum
 - $\uparrow$ **Contractility** $\Rightarrow$ $\uparrow$ stroke volume
- Much less heart rate response than other β_1 -agonists
- Mild vasodilatation from β_2 -adrenergic stimulation
 - $\downarrow$ Afterload
- Administered as continuous intravenous infusion
 - Used in acute decompensated heart failure

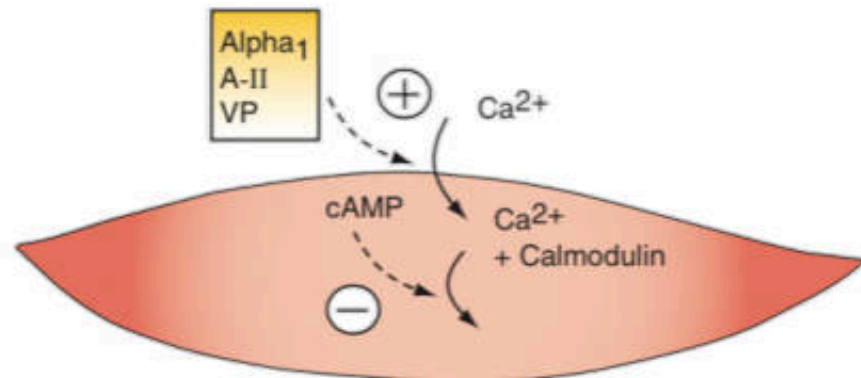


- $T_{1/2} = 2.5$ minutes
- **Adverse effects** (unusual at therapeutic doses)
 - Atrial/ventricular dysrhythmias
 - Precipitate myocardial ischemia

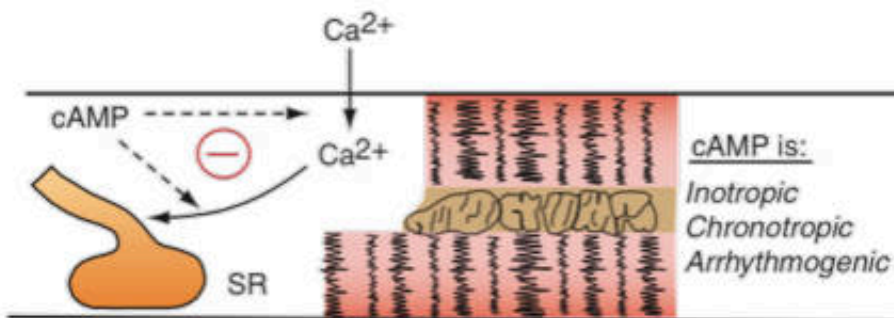
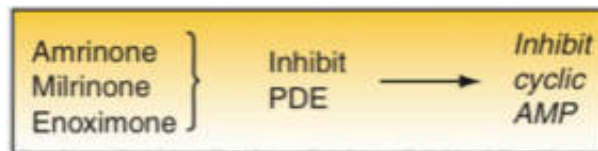
Phosphodiesterase Inhibitors: Milrinone

- Inhibits enzymatic hydrolysis of cAMP
 - $\uparrow$ cAMP $\Rightarrow$ activates protein kinases
 - $\uparrow$ Ca^{++} in sarcoplasmic reticulum
 - Effects heart and vasculature
- Clinical effects
 - $\uparrow$ Cardiac contractility $\Rightarrow$ $\uparrow$ stroke volume
 - Arterial/venous vasodilatation
 - $\downarrow$ Afterload & preload

Phosphodiesterase Inhibitors: Milrinone



ARTERIOLAR DILATION

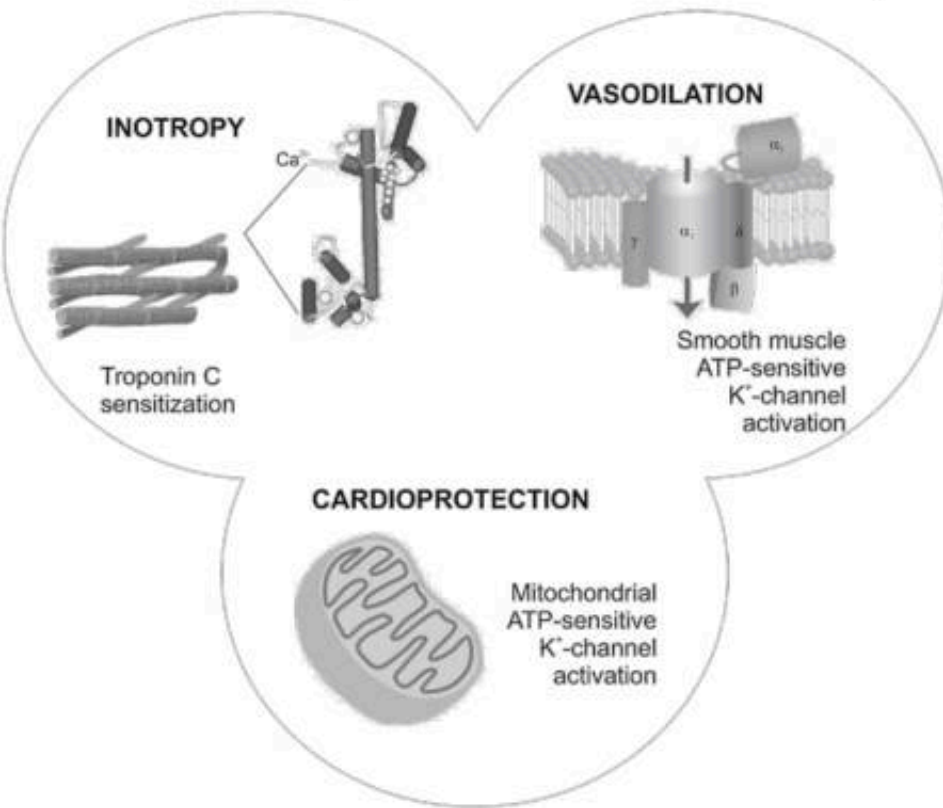


MYOCARDIAL CONTRACTION ↑

- Administered as constant IV infusion
 - Used in acute decompensated heart failure
- Adverse effects limit use
 - Hypotension
 - Thrombocytopenia
 - Dysrhythmias
 - May be associated with ↑ mortality

Levosimendan as an Inodilator

The effects of levosimendan, an **inodilator** indicated for the treatment of decompensated heart failure, are exerted via **Three pathways**

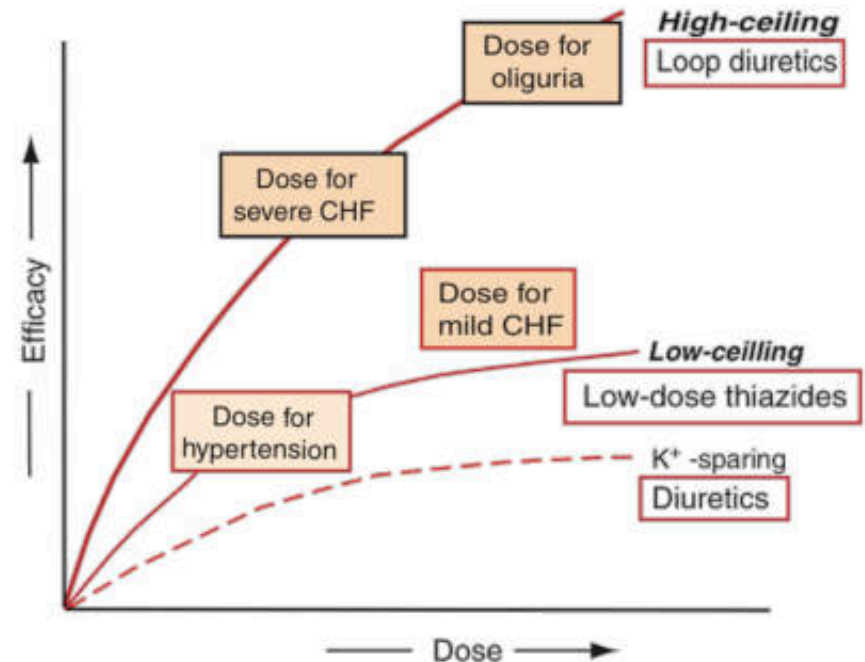
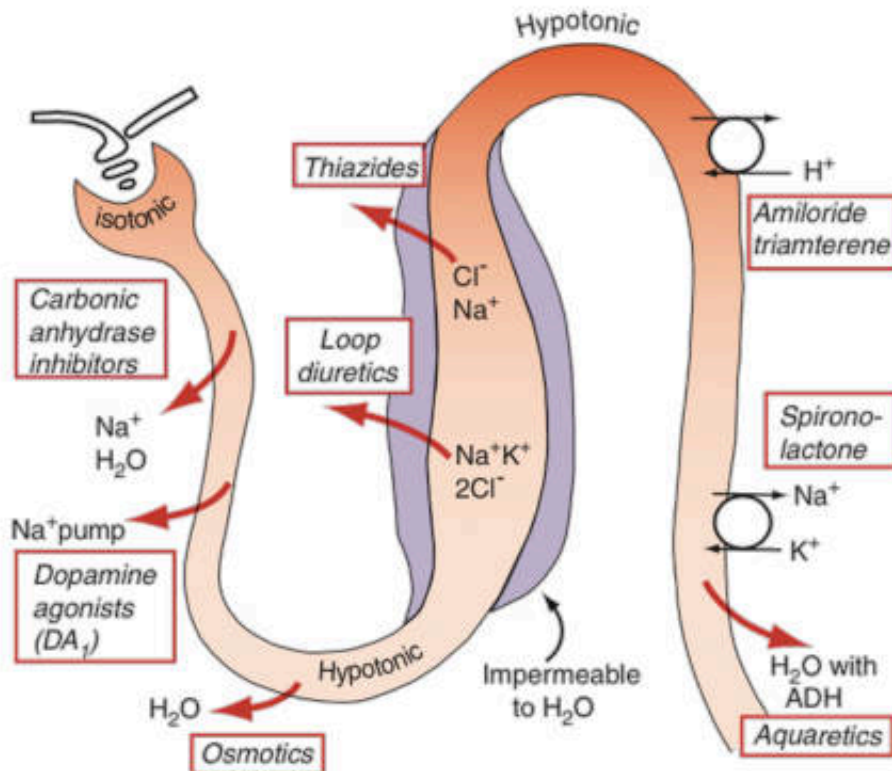


1. Increasing the sensitivity of cardiomyocyte troponin C fibers to ionic calcium; (**Inotropy**)
2. Opening of adenosine ATP-dependent potassium channels on vascular smooth muscle cells; (**Vasodilation**)
3. Opening of adenosine ATP-dependent potassium channels in mitochondria (**Cardioprotection**)

As it is a calcium sensitizer not a calcium mobilizer, levosimendan does not increase myocardial oxygen consumption, and prevents myocardial apoptosis and remodeling

Pharmacology of Volume Regulation

- **Diuretics:**
 - Drugs which increase the excretion of NaCl and water ↓ Preload
 - ↑ Volume of urine
 - Normally less than 1% filtered sodium is excreted in the urine

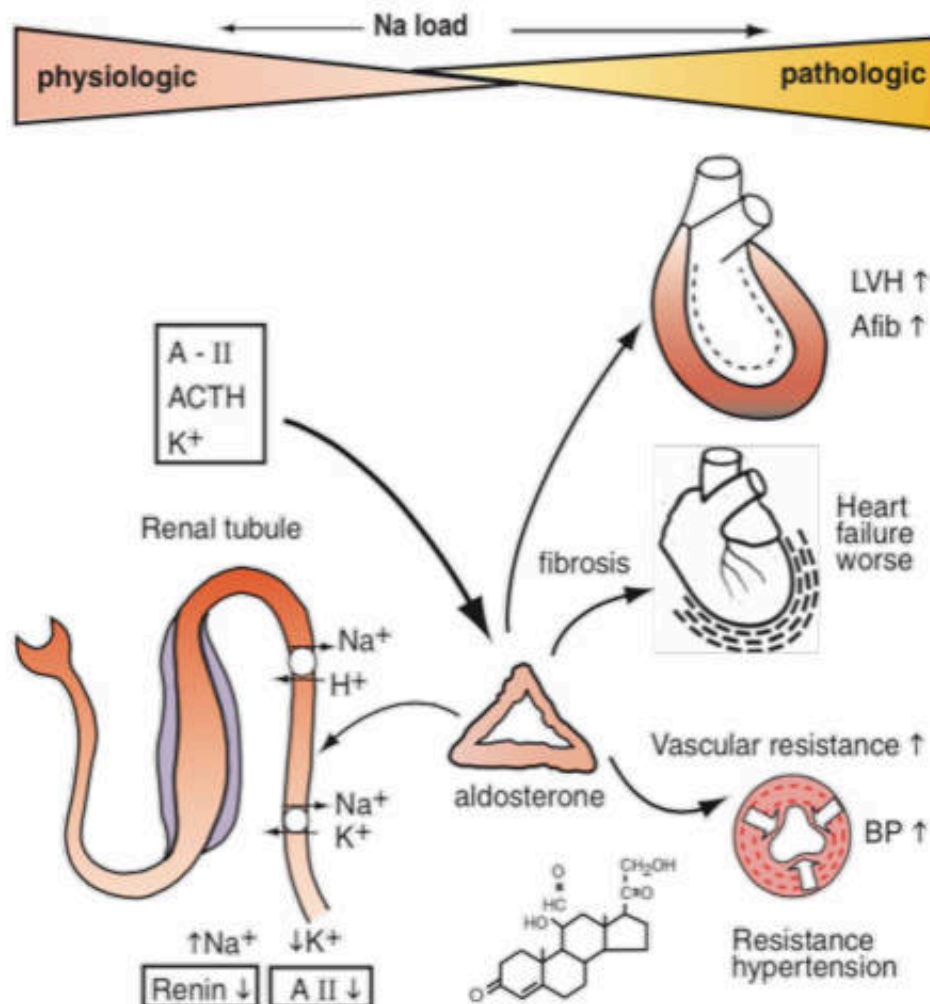


Loop Diuretics

- Mechanism:
 - Inhibit Na^+ - K^+ - 2Cl^- co-transporter of the ascending limb
 - Potent excretion of Na^+ & water
- Agents
 - Furosemide, bumetanide, torsemide, ethacrynic acid
- Front (First)-line therapy for heart failure & other edematous states
- Side effects:
 - Hypovolemia, Hypokalemia, Metabolic alkalosis, hypomagnesemia, Hyperuricemia, Hyperglycemia, Hyperlipidemia, Hyponatremia, Hypercalcemia, Ototoxicity – very high doses

Potassium Sparing Diuretics

- Weak diuretics act in collecting tubules
- **Spironolactone**
 - **Competitive antagonist of aldosterone**
 - Blocks aldosterone stimulated Na^+ reabsorption and K^+ and H^+ excretion in the late distal tubule and collecting duct
 - **Only diuretic to improve survival in NYHA Class III/IV**

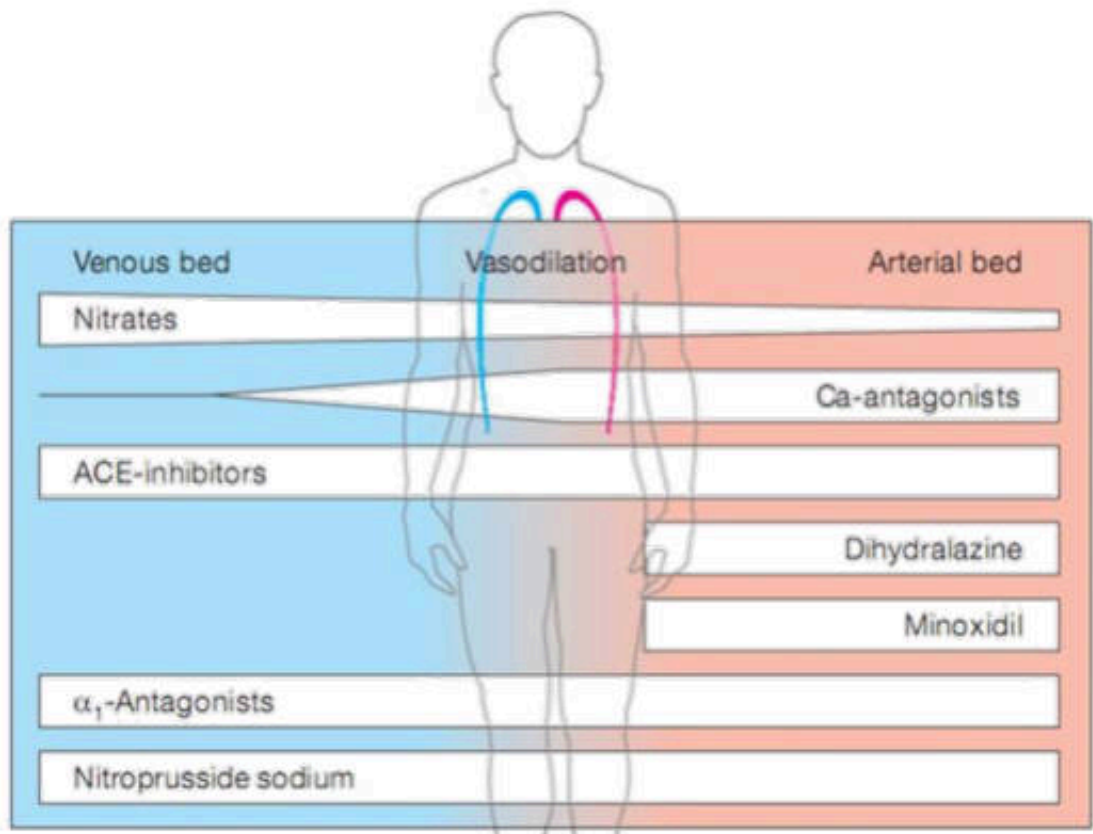


Potassium Sparing Diuretics

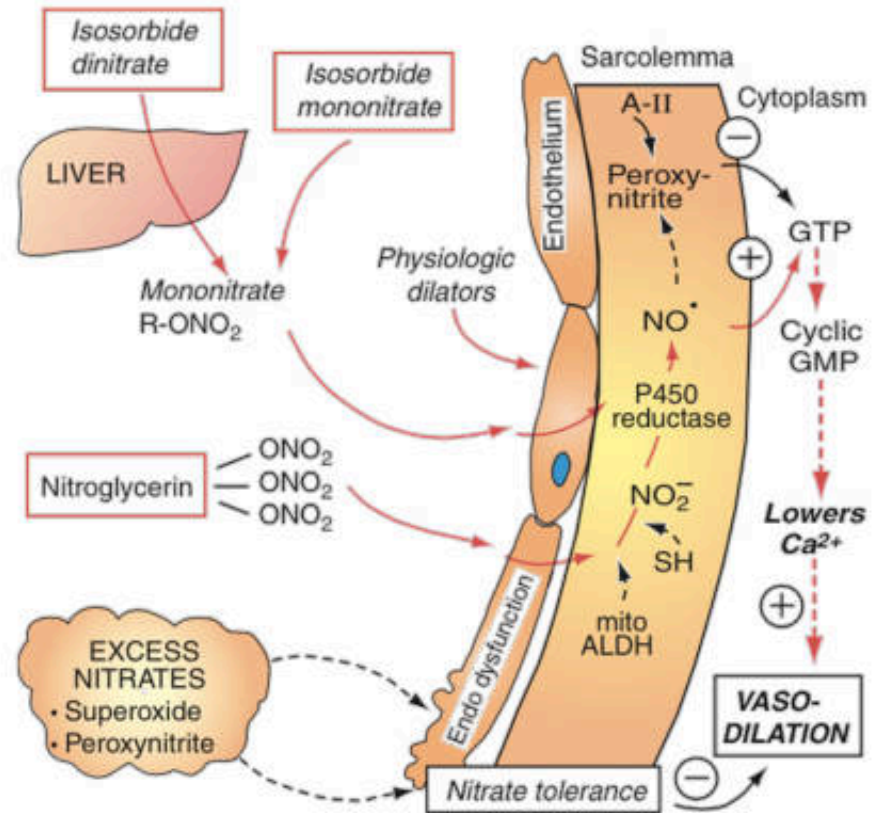
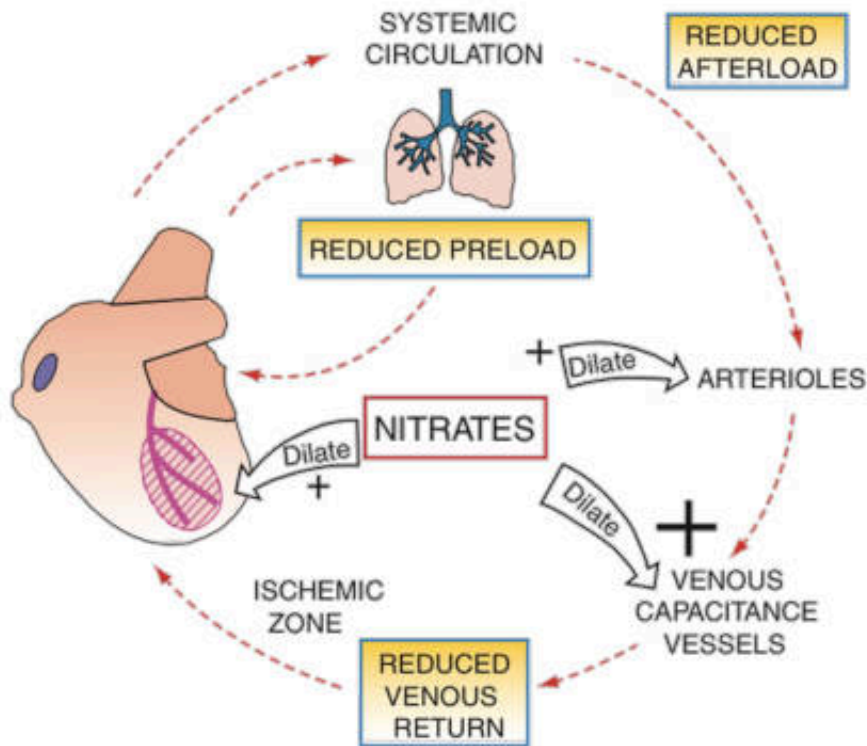
- Amiloride & Triamterene
 - Blocks Na^+ channels in late distal tubule and collecting duct.
 - K^+ and H^+ secretion into the urine is inhibited as this normally occurs in exchange for Na^+ .
 - Not associated with $\uparrow$ survival like spironolactone
- Side Effects for all K^+ sparing diuretics
 - Hyperkalemia
 - Metabolic acidosis

Pharmacology of Vascular Tone

- Donors of nitric oxide (NO)
- Angiotensin Converting Enzyme inhibitors (ACEI)
- Angiotensin Receptor Blockers (ARB)
- Other
 - Hydralazine
 - Nesiritide
 - β -Blockers



Donors of NO: Nitroglycerin



Donors of NO: Nitroglycerin

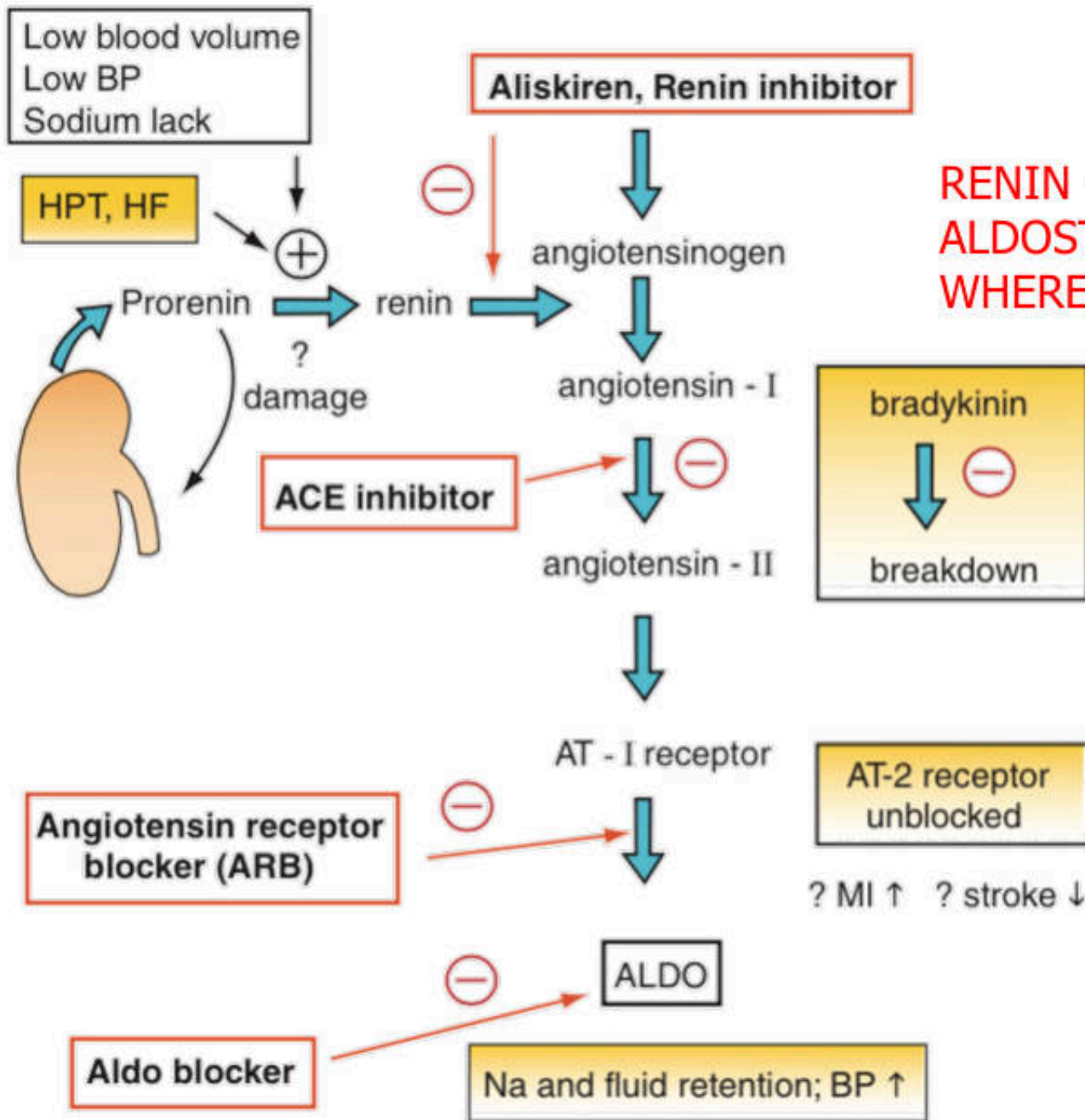
- Metabolized to nitric oxide (NO)
 - Results in vasodilatation
- Primarily venodilator at therapeutic doses
 - ↓ Preload
 - High doses $\Rightarrow$ venous & arterial vasodilatation
 - ↓ Preload & afterload

Nitroglycerin

- Available in intravenous, oral, topical, sublingual and buccal forms
- Nitroglycerin has short half-life
 - $T_{1/2} = 5$ minutes
- Oral forms (isosorbide dinitrate)
 - $T_{1/2} = 1 - 4$ hours
- Adverse effects
 - Headache – most common
 - Dizziness, lightheadedness
 - Hypotension

Donor of NO: Sodium Nitroprusside

- Vasodilatation by release of NO
 - Balanced arterial – venous vasodilator
 - ↓ Preload and afterload
- Only administered as continuous intravenous infusion
- $T_{1/2} = 5-10$ minutes
 - Onset of action = 2 minutes
- Metabolized to:
 - Cyanide $\Rightarrow$ thiocyanate
- Hypotension is most common adverse effect
 - Rarely cyanide poisoning can occur



RENIN - ANGIOTENSIN -
ALDOSTERONE SYSTEM:
WHERE INHIBITORS ACT

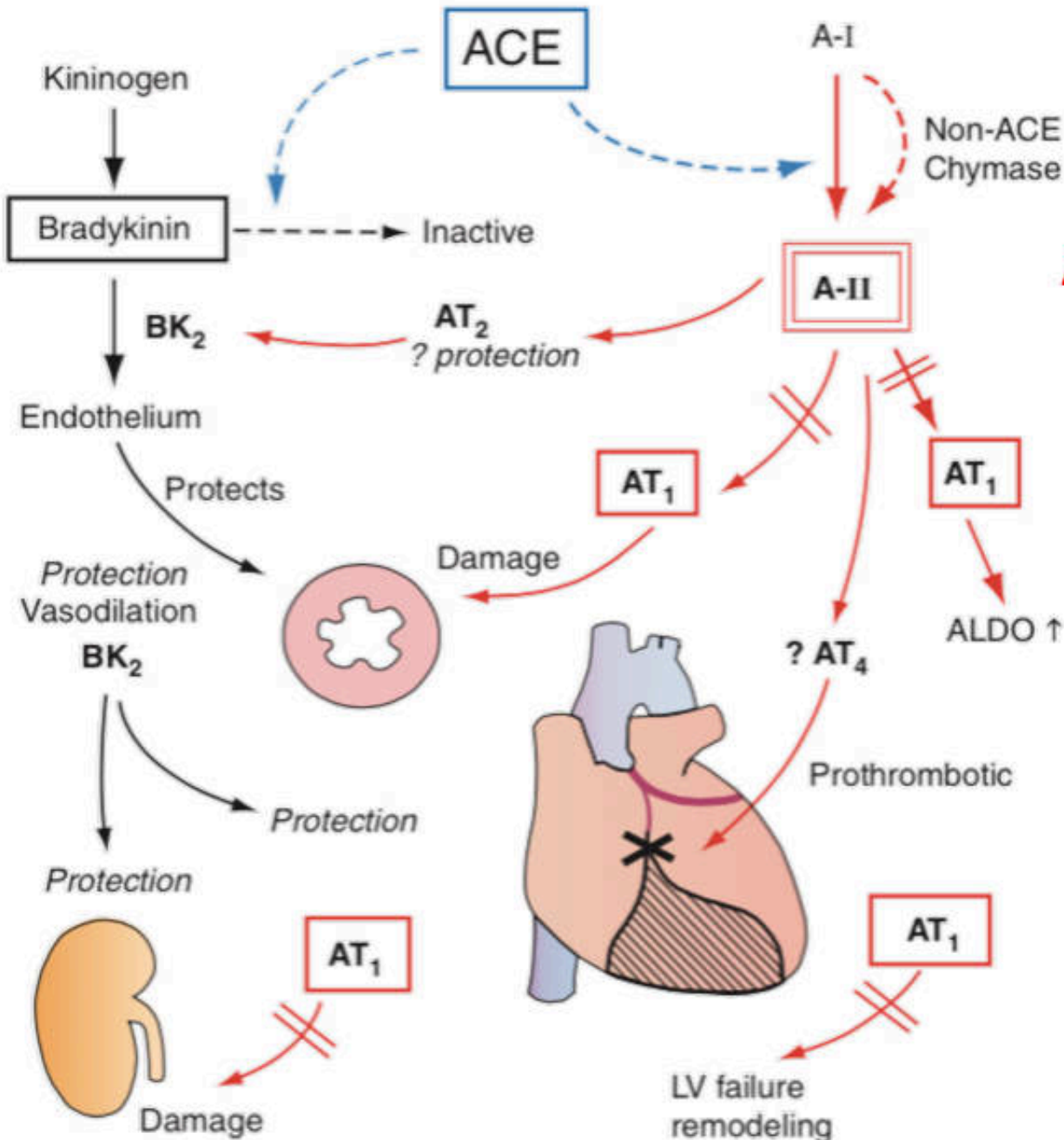
ACE

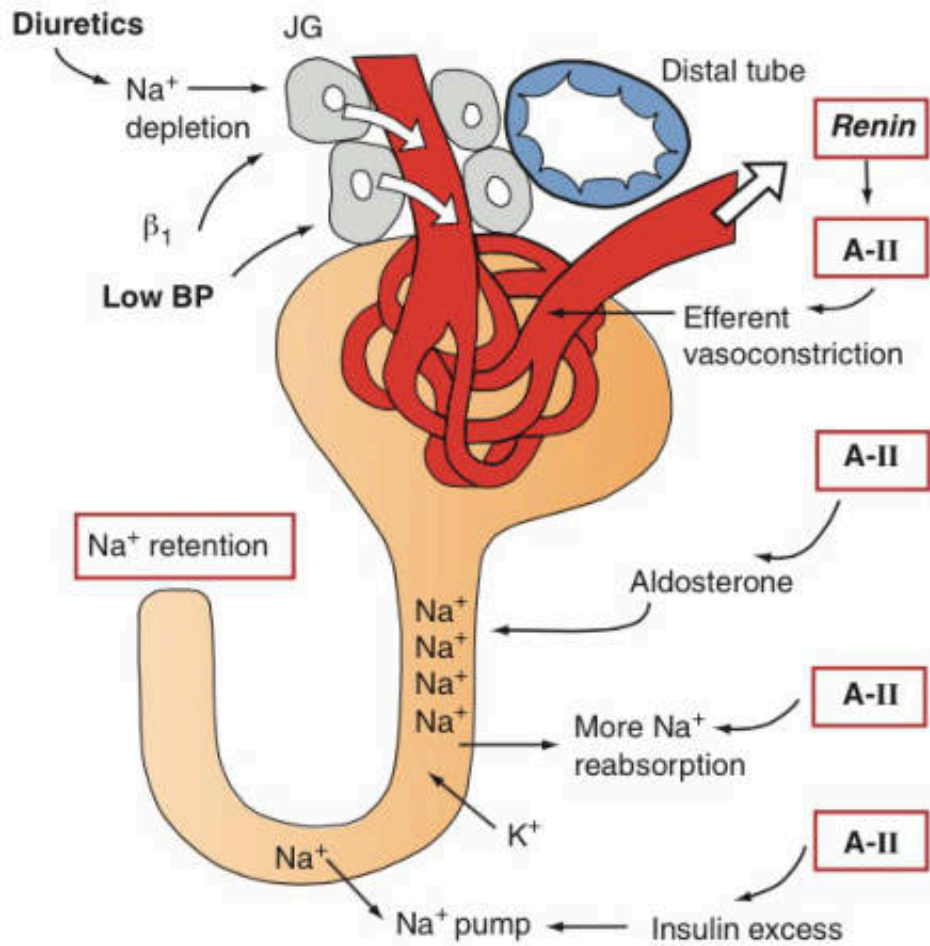
A-I

Non-ACE
Chymase

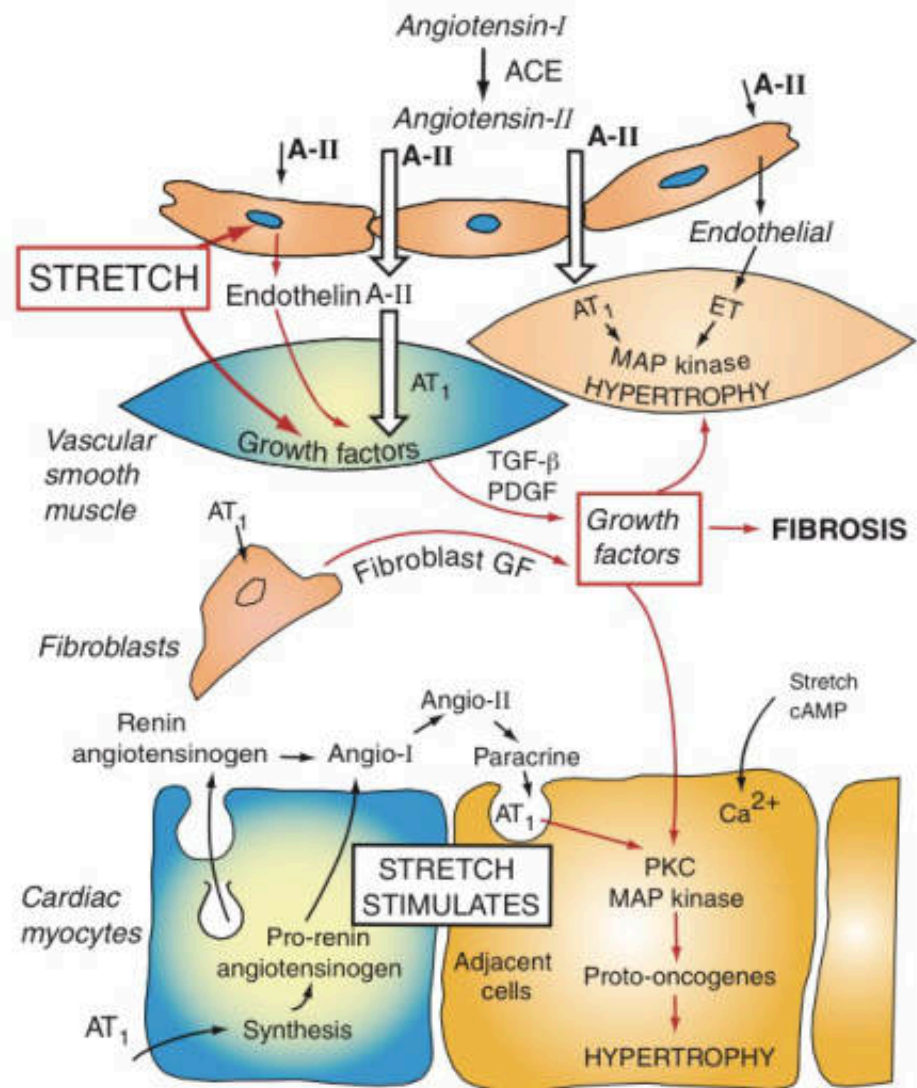
A-II

ACE, A-II EFFECTS and ARBs





A-II AND Na⁺ BALANCE



STRETCH AND CARDIAC GROWTH

ACEI

- Benzapril, Lisinopril, Captopril, Quinapril, Enalapril, Fosinopril, Ramipril, Trandolapril

Or any anything else with a “-pril”

- Inhibits generation of AII
 - ↓ Arteriolar vasoconstriction
 - ↓ NaCl reabsorption in the kidneys
 - ↓ Antidiuretic hormone release
- Inhibit bradykinin breakdown
- Results in arterial vasodilatation
 - ↓ **Afterload** & ↓ preload
- Improves survival in heart failure

ACEI	Starting Dose	Target Dose
Captopril	6.25 mg tid	50 mg tid
Enalapril	2.5 mg qd or bid	10 mg bid
Fosinopril	5.0 mg qd	40 mg qd
Lisinopril	5.0 mg qd	20 mg qd
Quinapril	5.0 mg qd	20 mg qd
Ramipril	1.25 mg bid	10 mg qd

ACEI: Adverse Events and Drug interaction

CNS: dizziness, fatigue, headache, insomnia, weakness

Resp: 20% develop dry cough,

CV: **hypotension**, angina, tachycardia

GI: taste disturbances, anorexia, diarrhea nausea

GU: proteinuria, impotence, **renal failure**

DERM: rash

HEM: Agranulocytosis, neutropenia (captopril only)

MISC: **Angioedema**; hyperkalemia

- ACEI $\Rightarrow$ most common cause of angioedema in adults

- **Drug interaction:** Potassium supplements, NSAIDs (Precipitate acute renal failure, Hyperkalemia), Potassium sparing diuretics (Hyperkalemia)

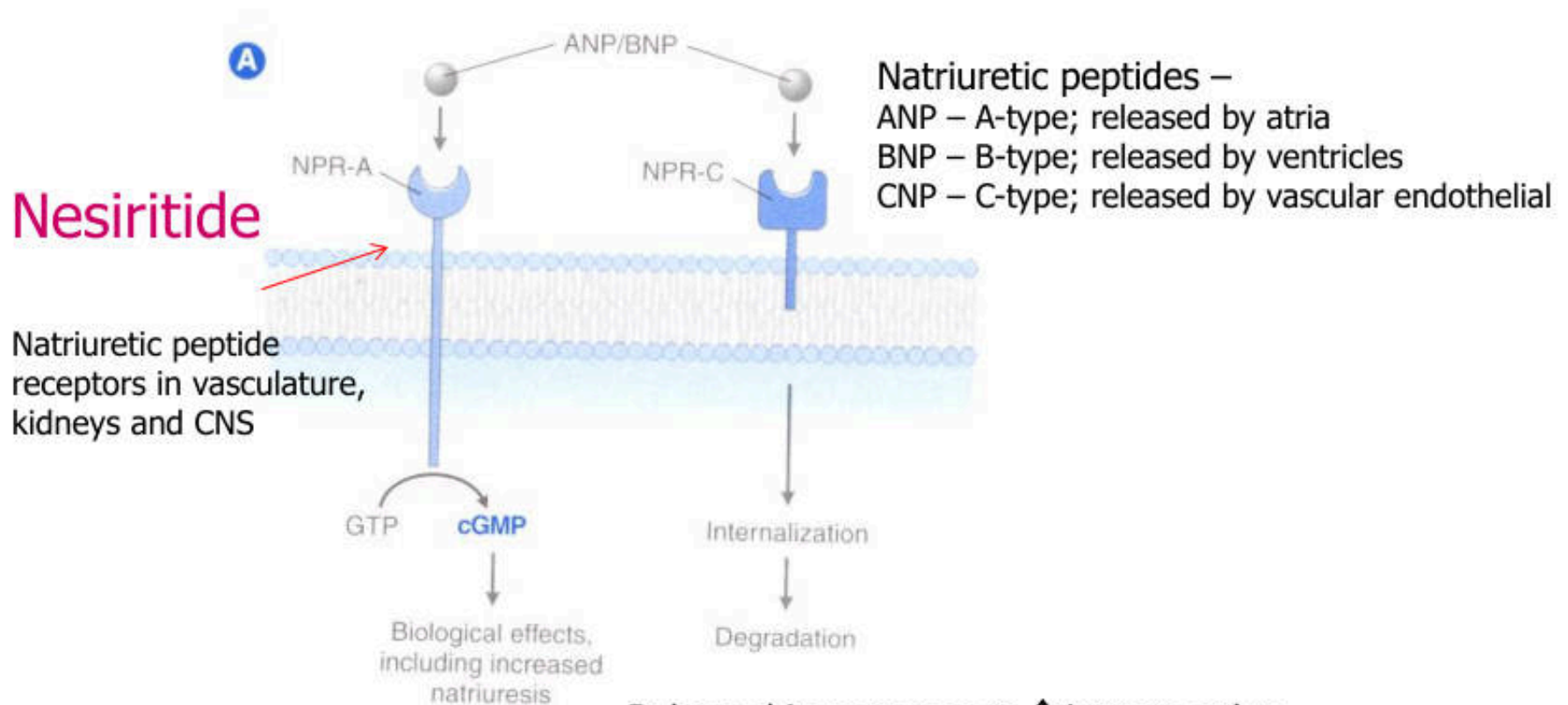
Angiotensin Receptor Blockers (ARB)

- Inhibit angiotensin II at its receptor
- No effect on bradykinin metabolism
- Similar effects/adverse effects as ACEI
 - Exception: unlikely to cause angioedema or cough
- **Losartan** (*Cozar[®]*), valsartan (*Diovan[®]*), cardesartan are the most commonly used ARBs
- May increase survival in heart failure
 - Usually not considered first-line drug

Hydralazine

- Arteriolar vasodilator
 - Mechanism unknown
 - ↓ **Afterload**
- Often used with nitrates (isosorbide dinitrate) in heart failure who cannot tolerate ACEI
- ↑ **Survival when used with nitrates**
 - Similar to ACEIs
- Adverse effects
 - Hypotension – most common
 - **Rare: drug-induced lupus erythematosus**

Physiology: Natriuretic Peptides



Results in:

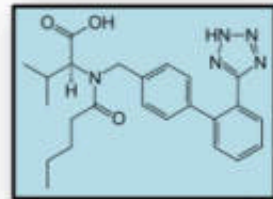
- ↑ Natriuresis/diuresis
- Venous & arterial vasodilatation
- ↓ Preload & afterload

Released in response to ↑ intravascular volume (atrial and/or ventricular dilation)

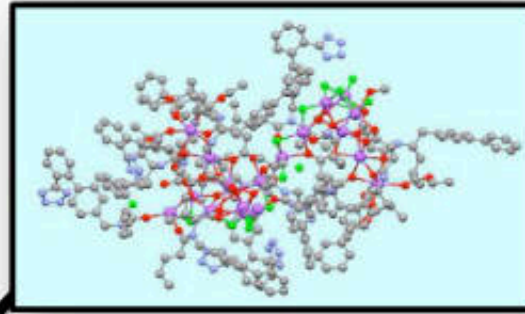
Nesiritide

- Recombinant BNP
- Administered only by continuous IV infusion
 - Used in acute decompensated heart failure
- Adverse effects
 - Hypotension - most common
 - May be associated with renal failure
 - May worsen mortality in heart failure

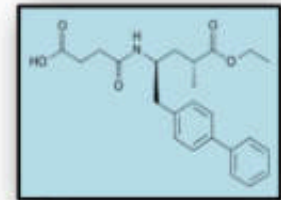
Valsartan/sacubitril, LCZ696: "Angiotensin Receptor-Neprilysin Inhibitor" (ARNi)



Valsartan



LCZ696



Sacubitril

Angiotensin
receptor blocker

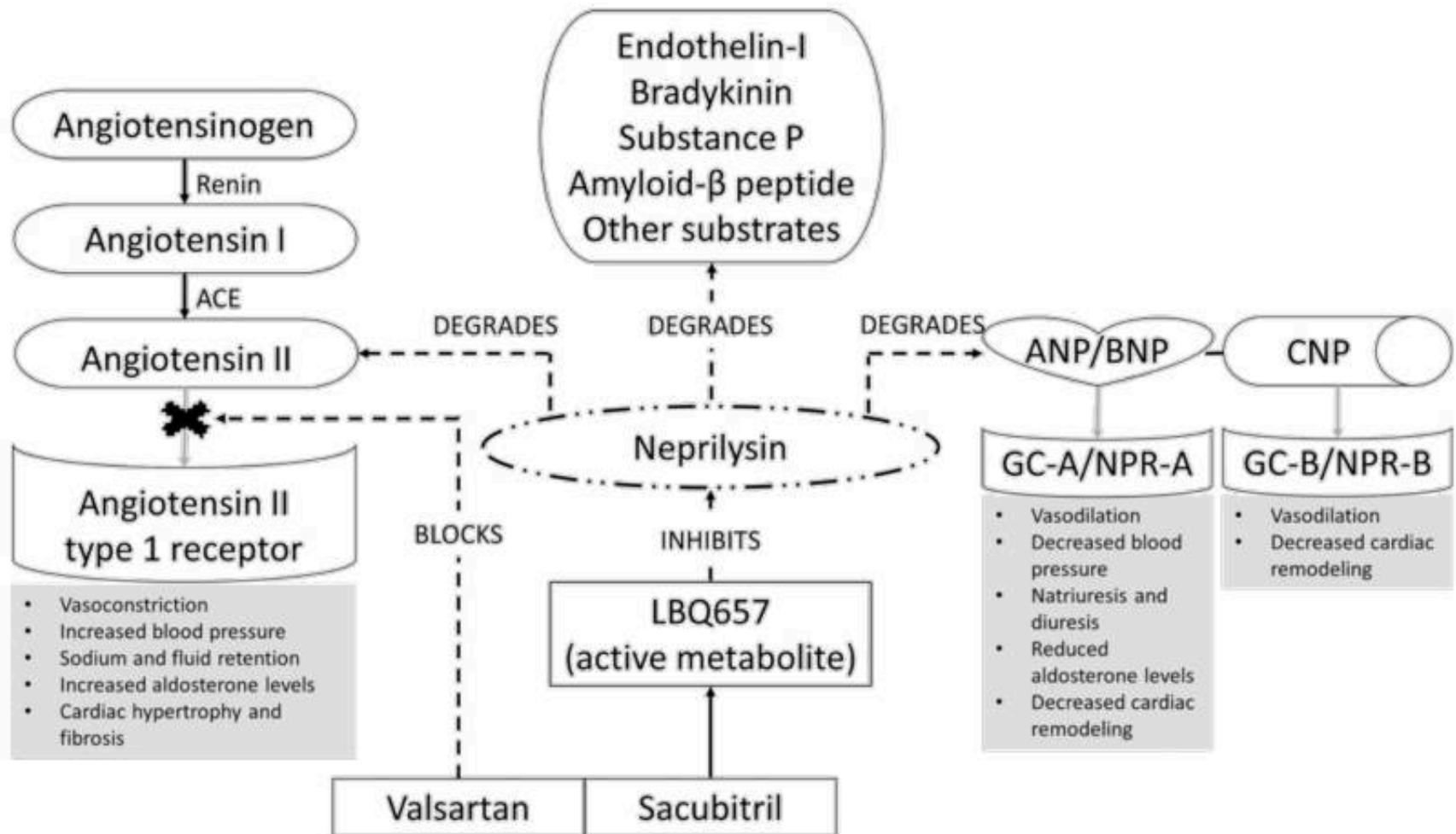


Inhibition of neprilysin

Increased activity or
response to maladaptive
mechanisms

Decreased activity or
response to adaptive
mechanisms

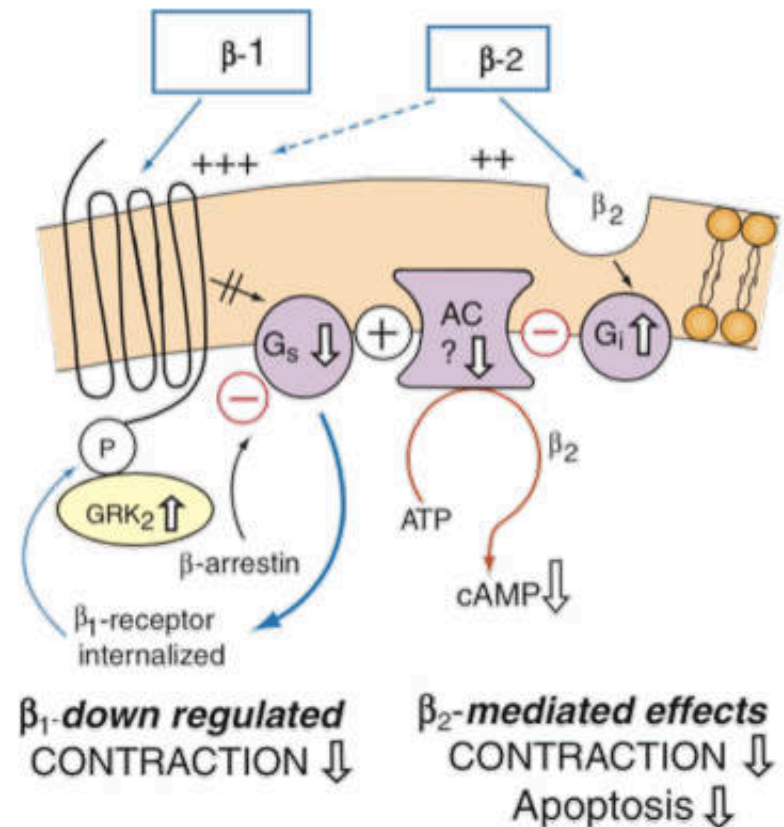
Angiotensin Receptor-Neprilysin Inhibitor" (ARNi)



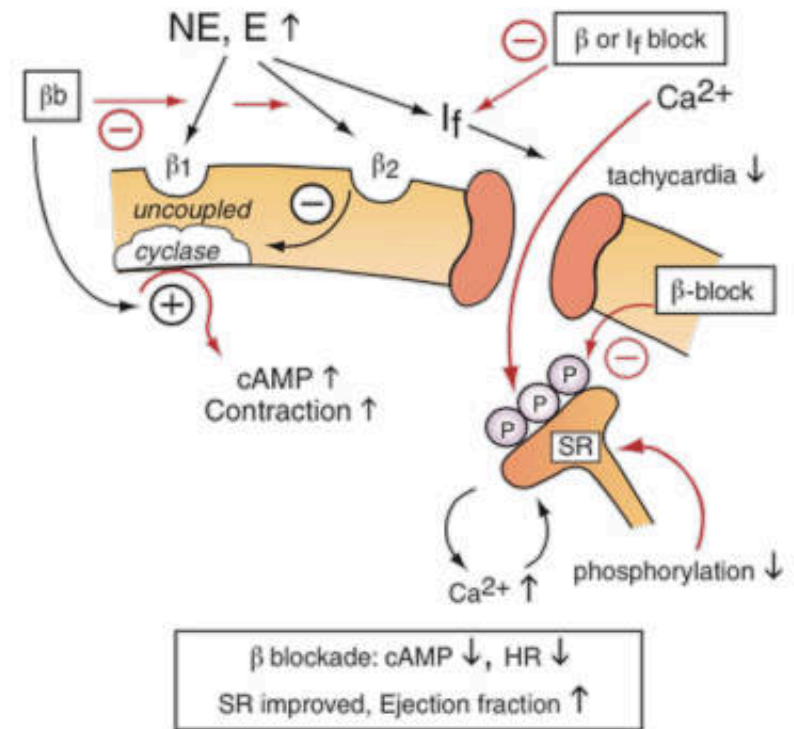
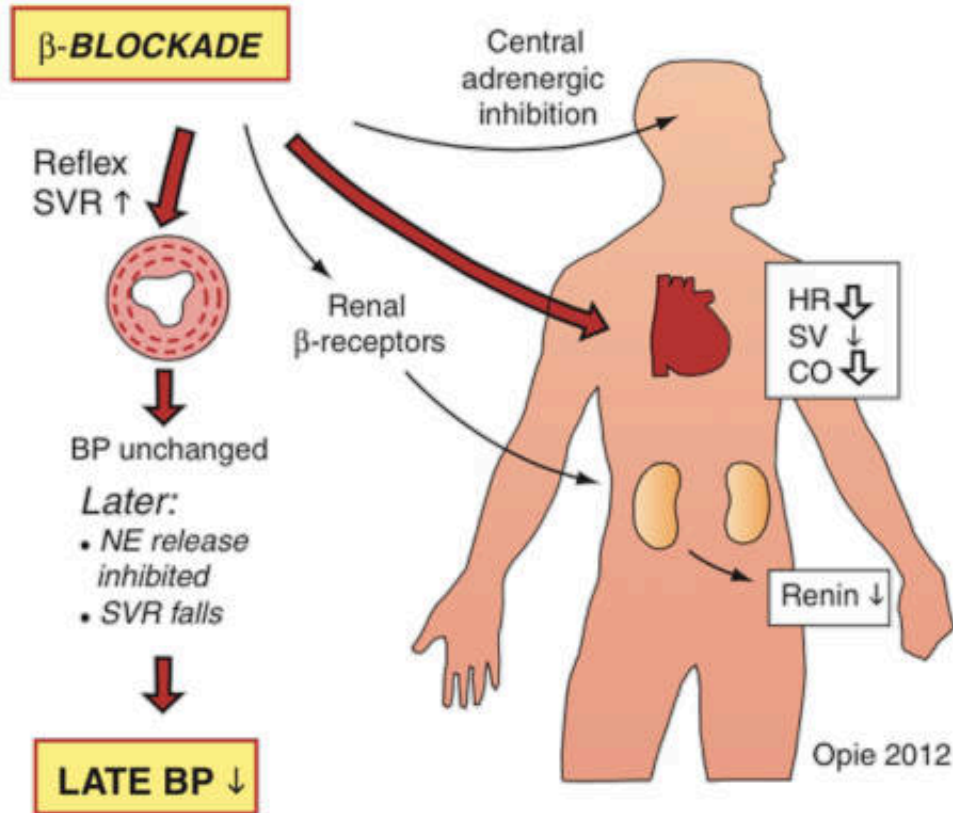
ACE, angiotensin-converting enzyme; ANP, atrial natriuretic peptide; BNP, brain natriuretic peptide; CNP, c-type natriuretic peptide; GC, guanylyl cyclase; NPR, natriuretic peptide receptor.

β -Adrenergic Blockers

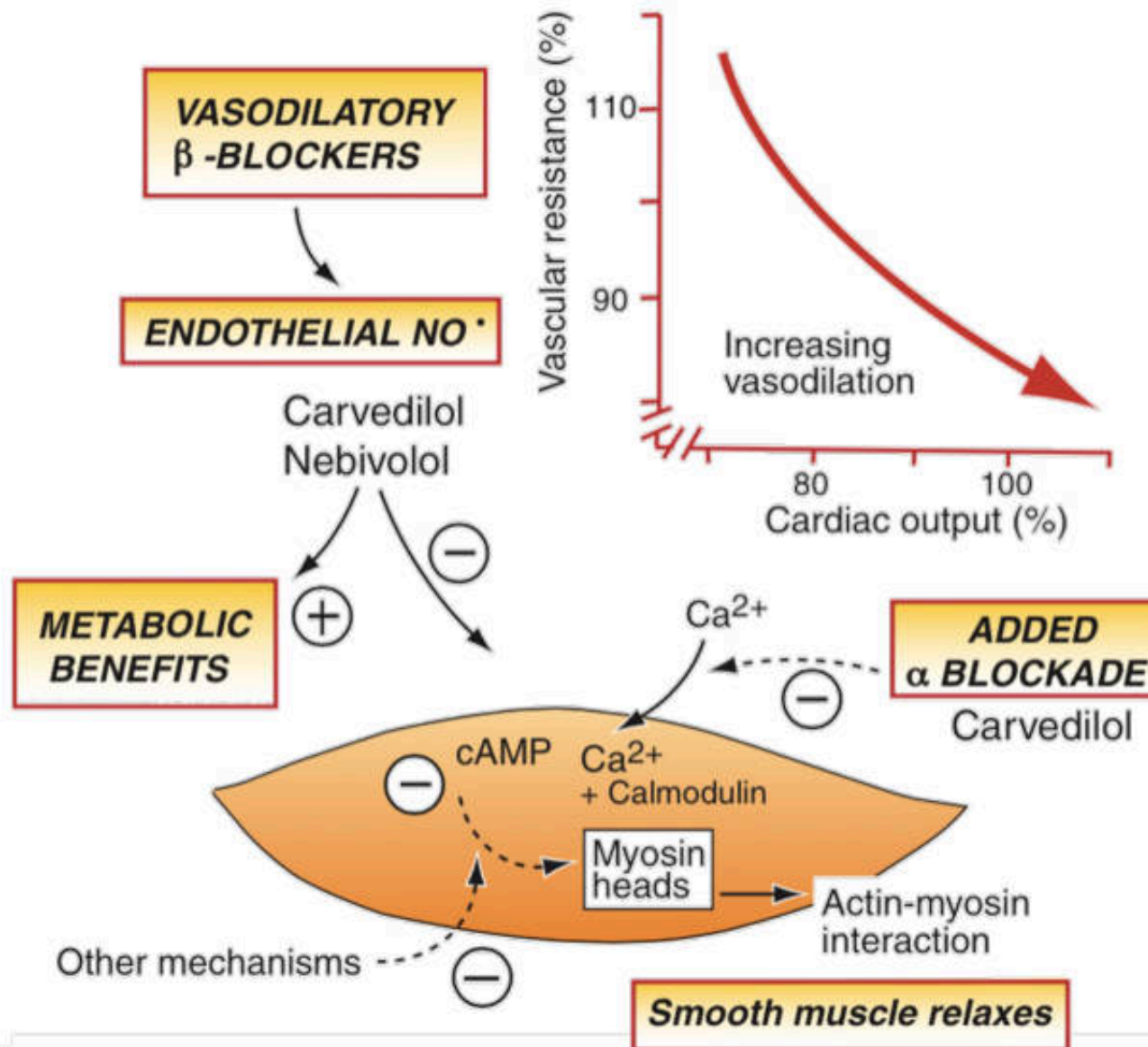
- Seems counterintuitive in HF
 - $\downarrow$ inotropy & chronotropy (afterload)
 - Inhibition of renin release (preload)
- Attenuation of cytotoxic & signaling effects of $\uparrow$ circulating catecholamines
 - Up-regulation of β_1 receptors
 - Restores receptor function
 - Improve myocardial re-modeling



β -Adrenergic Blockers



β -Adrenergic Blockers with vasodilator effects



β -Adrenergic Blockers

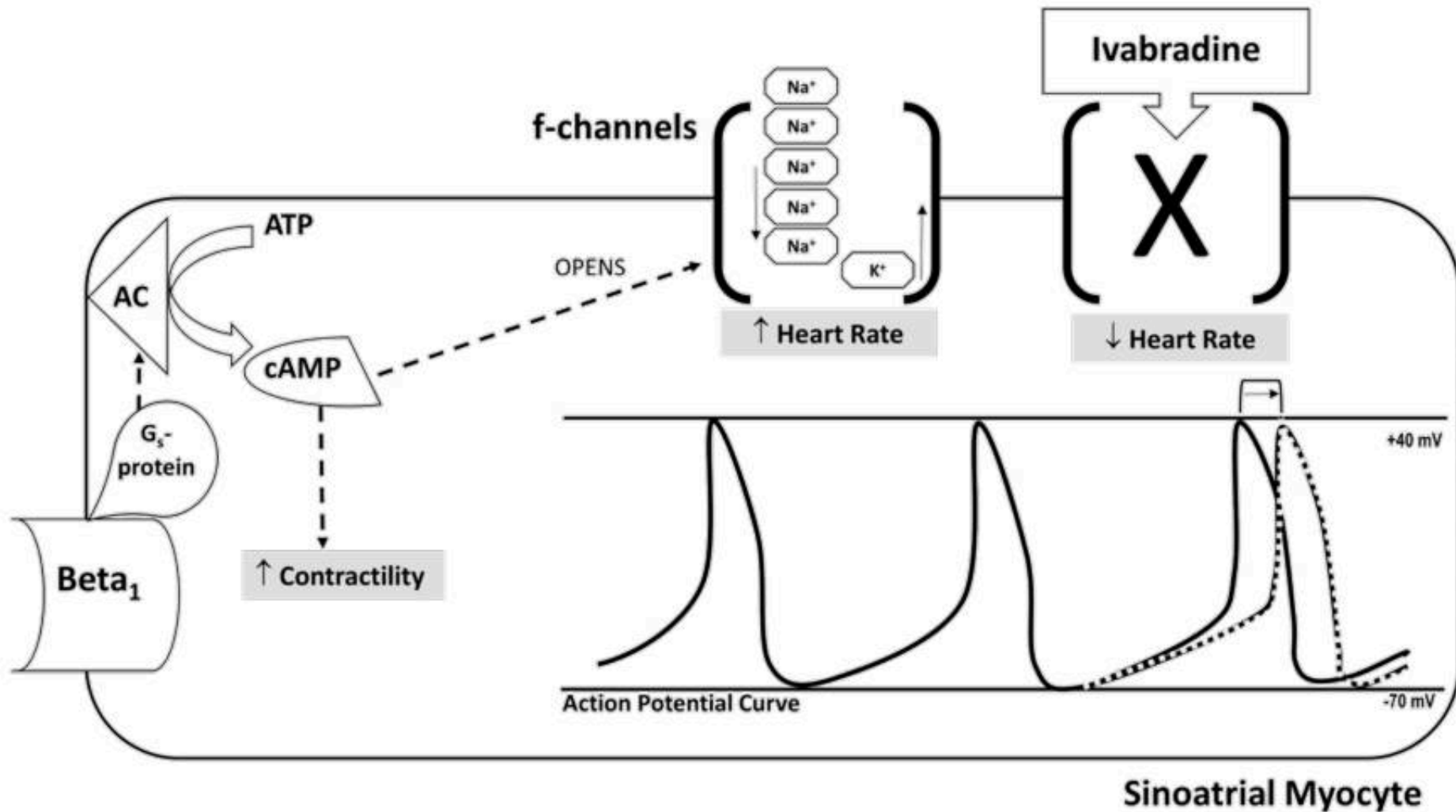
- Some β blockers are approved in HF
 - Metoprolol succinate - β_1 selective blocker
 - Bisoprolol - β_1 selective blocker
 - Nebivolol - β_1 selective blocker with NO-potentiating effect
 - Carvedilol - $\beta_1, \beta_2, \alpha_1$ blocker with NO-potentiating effect
- Carvedilol, metoprolol, bisoprolol and nebivolol associated with improved survival in heart failure
 - Survival benefit additive to that of ACEI
- Not used in acute therapy
 - Beneficial effects develop with long-term use

All are CYP2D6 substrate (bisoprolol – CYP3A4 and Carvedilol – CYP2C9)

Ivabradine

- For stable chronic HFrEF, LVEF $\leq 35\%$ and HR ≥ 70 at rest (sinus rhythm)
- **Ivabradine** selectively **inhibits** on the I_f (f is for "funny") in a dose-dependent manner, which is highly expressed in the **sinoatrial node**.
- I_f is a mixed $\text{Na}^+ - \text{K}^+$ inward current activated by hyperpolarization and modulated by the autonomic nervous system, which regulating pacemaker activity in the sinoatrial (SA) node.
- Blocking this channel **reduces cardiac pacemaker activity, slowing the heart rate** and allowing more time for blood to flow to the myocardium.

Ivabradine: Mechanism Of Action



Hyperpolarization-activated cyclic nucleotide-gated (HCN) channels in sino-atrial node

Other Pharmacological treatment options

- **Vericiguat** (LVEF <45%, NYHA class II to IV) is a novel soluble guanylate cyclase stimulator whose use may be considered in patients with a recent episode of worsening heart failure.
(vericiguat as a second-line treatment in patients who remain symptomatic despite first-line treatment, mainly after a recent decompensation)
- **Omecamtiv mecarbil** is a selective cardiac myosin activator whose further development has been halted by the manufacturer.
- **Iron deficiency** and skeletal muscle wasting have significant impact on patients' exercise capacity, quality of life and survival.

ENDOTHELIAL CELL

LAMINAR FLOW
AND SHEAR FORCES

Nitrates

eNOS

L-Arg

NO

O₂⁻

ONOO⁻

OXIDATIVE STRESS
NITROSATIVE STRESS

Comorbidities

Obesity
Hypertension
Diabetes mellitus
COPD
Iron deficiency
Renal insufficiency

CARDIOMYOCYTE

sGC stimulators
[eg vericiguat]

α β

sGC activators
[eg cinaciguat]

Trinitro-Bus
Troponin I
Phospholamban
ATP-sensitive potassium channel
L-type calcium channel
TRPC RGS2/4

GTP

PDE5

GMP

cGMP

Activates

PKG

PDE5 inhibitors
[eg sildenafil]

VASCULAR SMOOTH MUSCLE CELL

sGC stimulators
[eg vericiguat]

α β

sGC activators
[eg cinaciguat]

GTP

PDE5

GMP

cGMP

Activates

PKG

PDE5 inhibitors
[eg sildenafil]

L-type calcium channel
ATP-sensitive potassium channel
TRPC RGS2/4

FIBROBLAST

sGC stimulators
[eg vericiguat]

α β

sGC activators
[eg cinaciguat]

GTP

PDE5

GMP

cGMP

Activates

PKG

PDE5 inhibitors
[eg sildenafil]

HEART



Anti-hypertrophy
Anti-fibrosis
Improved coronary
blood flow
Anti-inflammatory

LV relaxation/compliance
Afterload/systemic
compliance
VA coupling

BLOOD VESSEL



Improved blood flow
Anti-proliferation/
remodelling
Anti-inflammation

Vasotone regulation

KIDNEY



Anti-fibrosis
Increased renal
blood flow

Renal protection

ADIPOSE TISSUE



Enhanced lipid uptake
Differentiation of
brown adipocytes

Anti-obesity effects